Unsupervised Domain Adaptation With Extended Multimodal LSTM for End-to-End Classification of Multisource Remote Sensing Data

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***Abstract*—Deep learning has gained success in unsupervised domain adaptation (UDA) for remote sensing (RS) data classification. However, existing methods lack specialized deep learning structure components for multisource RS (MSRS) data, restricting their performance. As such, a cross-modal compensation-consistent U-shape masked extended multimodal long short-term memory neural network (CCU-MaxNet) framework is proposed for UDA in end-to-end classification of MSRS data. First, the extended multimodal long short-term memory (xMLSTM) cell is designed as a new module, with which its down-block and up-block are built to explore the complementarity of MSRS data and long-term dependencies between their hierarchical features, serving as memory state. Then, a cross-modal differential compensation (CMDC) block is devised to enhance the complementary modality-specific information. Based on these blocks, a U-MaxNet is built to extract the representational features under the guidance of the memory state. A multimodal compensation consistent domain adaptation network (MCC-DANet) is further designed for domain alignment at both pixel and semantic levels. Finally, by integrating U-MaxNet and MCC-DANet, CCU-MaxNet extracts more discriminative domain-invariant features for cross-domain classification. Particularly, the proposed framework has excellent flexibility, providing a high-precision mode and a high-efficiency mode based on whether to use the fractional Fourier transform (FrFT) to meet the requirements of different applications. Extensive experiments on four cross-domain MSRS datasets verify its superiority compared to state-of-the-art methods. Our source code will be available at https://github.com/WenshuaiHu/UDA_MSRS_CCU-MaxNet**

***Index Terms*—Cross-modal differential compensation (CMDC), end-to-end classification, extended multimodal long short-term memory (xMLSTM), memory state, multimodal data generation, multisource remote sensing (MSRS) data, unsupervised domain adaptation (UDA).**

I. Introduction

THE continuous developments of remote sensing (RS) technology provide us with unlimited possibilities to obtain large-scale multisource RS (MSRS) data for Earth observation (EO), such as multispectral images (MSIs), hyperspectral images (HSIs), synthetic aperture radar (SAR), and light detection and ranging (LiDAR) data [1], [2]. This promotes their applications in various tasks, such as precision agriculture [3], urban planning [4], [5], wetland monitoring [6], [7], and military reconnaissance [8]. In recent years, the emergence of deep learning (DL) provides these tasks with technical support [9], where various DL-based models are built for HSI classification [10], [11], [12], [13], and MSRS data classification [14], [15], [16], effectively improving the classification accuracy.

The above methods usually require high-quality labeled data and are often designed based on the same distribution assumption of training (source domain) and testing data (target domain). However, large-scale annotations of RS data, especially MSRS data, are laborious and expensive in practice [1], [17]. Affected by various factors (i.e., sensor type, diurnal variation, and seasonal change), there are significant distribution differences between source and target domains, which lead to spectral shifts [18] and spatial texture changes [19] of the same type of land covers between different scenes or the same scenes at different times, yielding unsatisfactory performance. The above phenomenon is called domain shift [20], and has promoted research on domain adaptation (DA) [21], forming various feature-, domain-, or class-level DA technologies to achieve satisfactory results in the field of computer vision [22], [23].

Subsequently, the DA-based methods have been introduced into the field of RS [24]. Different from cross-domain HSI classification tasks [25], [26], [27], the cross-domain MSRS data classification task contains complex distribution differences in data between domains, where multidimensional, multimodal, and multidomain information increase the difficulty of solving such a task. Zhao et al. [28] first analyzed this task, and proposed a fractional fusion and spatial–spectral DA (FrF-SSDA) to overcome the spectral shift of HSIs

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using LiDAR data. This cross-domain classification task was defined in [29], forming a multilevel DA network (MDA-NET) to improve the classification accuracy in the target domain. A semisupervised cross-domain feature fusion classification network was designed for wetland classification, with unsupervised cross-domain feature fusion and supervised classification [7]. Yang et al. [1] built a graph embedding interclass relation-aware adaptive network, where the aligned features are used to perceive interclass relationships for class knowledge transfer between domains. To address the unpredictable feature distributions due to the lack of interpretability in the deep learning methods, a contrastive constrained cross-scene model-informed interpretable classification strategy (C^3MI-C) was built [30]. From the perspective of feature decoupling, a shared-private decoupling-based multilevel feature alignment semisupervised (SASS) was built [31]. To promote the development of semantic segmentation, a cross-city semantic segmentation (C2Seg) contest was organized by the IEEE Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing (WHISPERS) in 2023, releasing a new MSRS benchmark containing HSI, MSI, and SAR images [32]. Some C2Seg methods have been devised, such as a high-resolution DA network (HighDAN) [33].

Although the above methods overcome the distribution differences of MSRS data between different scenes, there are still some tough issues. First, they are typically designed using existing deep learning components (i.e., convolution, long short-term memory (LSTM), graph convolution, and transformer), which were originally designed for unimodal data, lacking dedicated multimodal components. Second, most existing methods usually have a high demand for high-quality labeled data. In recent years, as a self-supervised learning paradigm, mask image modeling (MIM) has received widespread attention in the field of computer vision, demonstrating advantages in perceiving spatial details and learning fine-grained representations from unlabeled data [34], [35], [36]. It may be more likely to be suitable for MSRS data classification due to the lack of sufficient labeled samples [26], [37], [38], [39]. However, existing MIM-based methods were designed using traditional mask autoencoders (MAEs) [34], and overlooked the utilization of scale information and intralayer and interlayer relevant information, limiting their performance [14], [40]. Finally, most of the existing cross-domain MSRS data classification methods are patch-based, and usually require pixel-by-pixel processing with high complexity, limiting their practical applications, especially for large-scale scenes [33], [39], [41].

As such, a cross-modal compensation-consistent U-shape masked extended multimodal LSTM (xMLSTM) neural network (CCU-MaxNet) is proposed for unsupervised DA (UDA) in end-to-end classification of MSRS data. The contributions mainly lie in the following four aspects.

1) The xMLSTM cell is designed as a new deep learning structure component to model the dependency between MSRS data or their hierarchical features. This dependency is defined as its memory state to guide multimodal feature extraction, and can be integrated into the forward propagation process of deep networks without introducing additional complexity.
2) xMLSTM down-block and up-block are built as two plug-and-play blocks to design a hierarchical encoder–decoder backbone, and a cross-modal differential compensation (CMDC) block is devised to explore and enhance the modality-specific complementarity. They are compactly integrated to design a U-MaxNet, which can extract complementarity-enhanced hierarchical features and their fusion representations.
3) A multimodal compensation consistent DA network (MCC-DANet) is built to eliminate domain gaps between MSRS data from different domains and enable U-MaxNet to adapt to the target domain through pixel- and semantic-level domain alignments under the cycle-consistent learning mechanism and the memory state.
4) A CCU-MaxNet framework is developed by integrating U-MaxNet and MCC-DANet to extract more discriminative domain-invariant representations, improving the cross-domain classification performance. Moreover, it offers two modes with good flexibility: a high-precision mode and a highly efficient mode, depending on whether the fractional Fourier transform (FrFT) is introduced. This allows for a tradeoff between cost and precision according to practical application requirements.

The rest of this article is organized as follows. Section II focuses on the related works. Section III elaborates on the proposed CCU-MaxNet. The experimental results with analysis are given in Section IV. Conclusions are presented in Section V.

II. Related Work

A. Typical Deep Learning Components for Feature Extraction in RS

With the development of deep learning, various structural components have been designed and introduced in the field of RS. The convolutional (Conv) layer was the core component for designing various convolutional neural networks (CNNs), such as 1-D CNN [10], 2-D and 3-D CNNs [11], and HighDAN [33]. To enhance their ability to learn spatial relationships or interclass dependencies, graph networks [1], [16], [29], [42], and capsule networks [43], [44] were devised. From the perspective of modeling long-range correlations in sequence data, LSTM, a special recurrent cell, was designed [45] and extended to various variants, such as 2-D Conv LSTM (ConvLSTM2D) [46], ConvLSTM3D [13], tensor-train ConvLSTM2D [47], and pseudo complex-valued ConvLSTM2D [48]. Recently, aiming at poor performance of the above components in processing long sequence data, some new components, such as transformer [49], [50], [51], [52], and Mamba [53], [54], extended LSTM (xLSTM) [55], were developed to effectively improve parallel processing capability.

The above components were originally designed for unimodal data, and cannot directly model the correlations between modalities and between layers. Although researchers have tried to design some multimodal deep learning components [56], [57], multimodal data were still fused with weighted summa-

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tion, failing to use their complementarity. In this article, a new multimodal component (xMLSTM) is designed to model the long-range correlations between hierarchical features through a special memory state for the classification of MSRS data.

B. Cross-Domain MSRS Data Classification

By integrating these components and DA strategies, various methods have been proposed for cross-domain MSRS data classification. With the adversarial learning, the data-level fractional fusion and feature-level adversarial alignment were used to reduce feature disparity [28]; HighDAN was devised for C2Seg [33]; HMDA-DANet was the first to design a multimodal MAE that can be trained from scratch, achieving more efficient and high-performance cross-domain classification [39]. With the convolution and improved transformer, a multimodal feature fusion and fine-tuning strategy was built for semisupervised cross-domain classification [7]. Dong et al. [30] designed a C^3MI-C to explore a shared subspace for multimodal domain alignment. Based on the convolution and graph components, Zhang et al. [29] introduced optimal transport and heterogeneous knowledge graphs matching to enhance domain alignment of topological structures. A graph-based progressive hierarchical feature extraction network was built, followed by joint de-scrambling alignment and pseudolabel generation for cross-scene classification [1]. With the convolution, transformer, and graph components, an SASS model was proposed for pseudolabel generation and cross-domain classification [31].

Most of them are supervised and patch-based, which heavily rely on labeled samples and ignore multimodal relationships for DA, limiting their performance and efficiency. In this article, a multimodal differential compensation-guided MCC-DANet is devised for pixel- and semantic-level distribution alignment.

C. Mask Image Modeling

Different from the above supervised methods, MIM, as a self-supervised learning paradigm, focused on utilizing only some visible parts to reconstruct original inputs, forming various generative models to explore valuable representations from unlabeled data by understanding their structures [34], [35]. Considering that obtaining high-quality labeled samples is time-consuming and laborious, MIM-based methods have been developed in the field of RS [37], [38]. For MSRS data, based on their fusion results with concatenation, Lin et al. [58] designed a transformer-based spatial–spectral MAE (SS-MAE) with independent spatialwise and spectralwise branches for spatial–spectral reconstruction. Li et al. [59] introduced the cross-attention and contrastive learning to enhance the interaction between these two branches for spatial and spectral feature fusion. A multimodal unified representation learning framework was designed, where a multistage preselected sensor-specific feature extractor and vision transformer are designed, followed by single-modal and multimodal fine-tuning for hierarchical representation and classification [60].

The above methods usually use existing large-scale multimodal datasets or a subset of raw MSRS data for pretraining, and fail to fully consider or explore their inherent characteristics (i.e., multiscale information, multimodal complementarity, and domain distribution discrepancy). In this article, we design a new U-MaxNet to extract rich multimodal multiscale features through long-term correlation modeling, hierarchical difference compensation, and cross-modal reconstruction, effectively improving classification accuracy of MSRS data.

III. Proposed CCU-MaxNet Framework

A. Architecture Overview

Fig. 1 shows the structure of the proposed CCU-MaxNet. For notational clarity, the symbols are introduced in detail. Suppose that MSRS data in the labeled source domain are $\mathcal{S} = \{(\mathcal{X}_H^s, \mathcal{X}_L^s), Y^s\}$, where Y^s is the known source label space with N classes; while $\mathcal{T} = \{(\mathcal{X}_H^t, \mathcal{X}_L^t)\}$ is the unlabelled target domain, where the target label space Y^t is unknown. s and t mean the source and target domains, $\mathcal{X}_H \in \mathbb{R}^{k\times k\times c_h}$ and $\mathcal{X}_L \in \mathbb{R}^{k\times k\times c_l}$ are HSI and LiDAR data, with image size $k \times k$ and channel numbers c_h and c_l, respectively.

B. Extended Multimodal LSTM (xMLSTM)

Inspired by the the sequence processing capability of transformers [39], [51], [52] and the gating mechanism of LSTMs [13], [45], [46], [47], [48], a novel xMLSTM cell is designed for MSRS data, with which its down-block and up-block are devised as plug-and-play blocks for design of encoder–decoder networks, extracting multimodal hierarchical features by modeling their long-range correlations as the memory state. Its update process can be well integrated into the optimization process of networks without increasing too much complexity. Then, a CMDC block is built to enhance modality-specific intrinsicality and cross-modal difference information. Finally, a new U-shape hierarchical network (namely U-MaxNet) is devised to overcome the influences of heterogeneity and extract more representative fusion features for better classification of MSRS data.

1) xMLSTM Cell: The xMLSTM cell is designed by introducing two new gates and a new memory state for multimodal complementarity interaction and feature extraction. First, by integrating the original input and forget gates, a new filter gate is devised to explore long-range complementary correlations between multimodal data as the memory state. Then, two selection gates are developed to update the multimodal features with the above memory state. Fig. 2 depicts the structure of the xMLSTM cell, whose formulas are defined as

$$\begin{aligned} q_c, k_x, v_x &= f_{fc-q}\left(f_{cc}\left(C_{l-1}\right)\right), f_{fc-k}\left(X_l^1\right), f_{fc-v}\left(X_l^2\right) \\ F_l &= \sigma\left(f_{fc-c}\left(f_{ca}\left(q_c, k_x, v_x\right)\right)\right) \\ C_l &= F_l \circ \left(tanh\left(f_{fc-c}\left(f_{ca}\left(q_c, k_x, v_x\right)\right)\right) + C_{l-1}\right) \\ H_l^1 &= \sigma\left(X_l^1\right) \circ \tanh\left(f_{c-1}\left(C_l\right)\right) \\ H_l^2 &= \sigma\left(X_l^2\right) \circ \tanh\left(f_{c-2}\left(C_l\right)\right) \end{aligned} \tag{1}$$

where X_l^1 and X_l^2 are multimodal inputs in the current layer, H_l^1 and H_l^2 are the outputs, and C_{l-1} and C_l are memory states to measure long-range dependencies between multimodal features at different layers. $f_{fc-q}(\cdot)$, $f_{fc-k}(\cdot)$, $f_{fc-v}(\cdot)$, $f_{fc-c}(\cdot)$, $f_{fc-1}(\cdot)$, and $f_{fc-2}(\cdot)$ are the fully connected (FC) layers with

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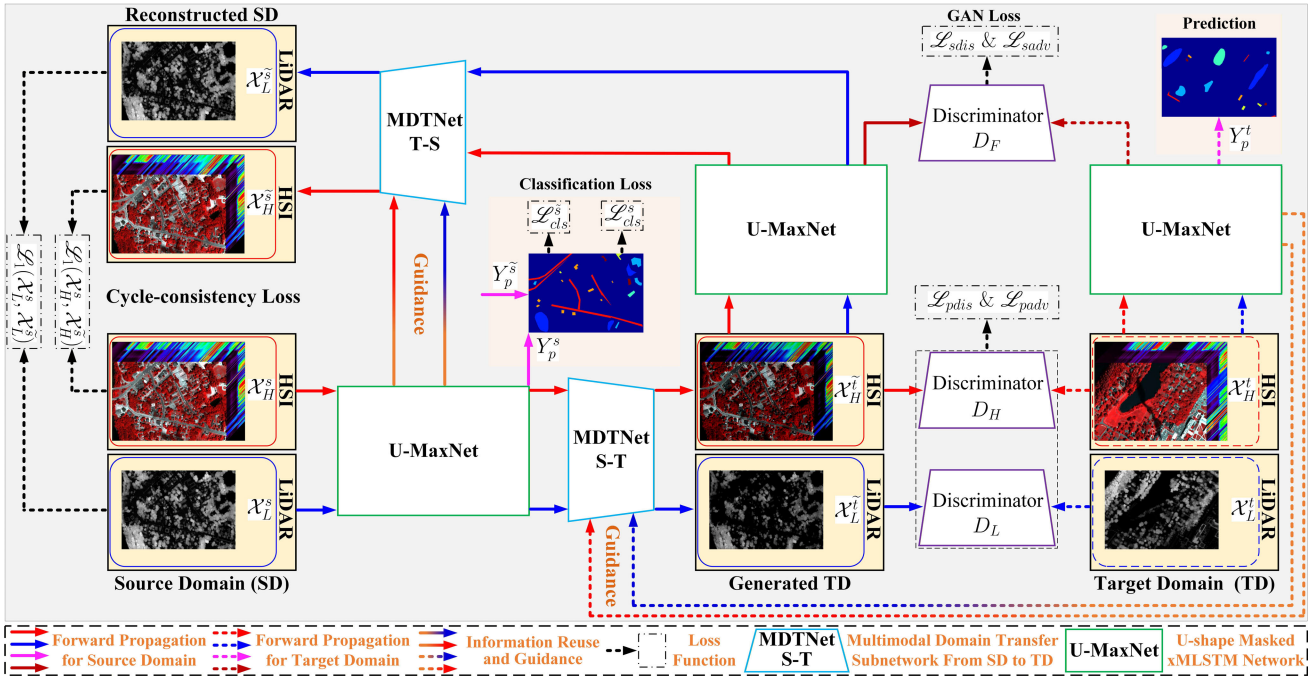


Fig. 1. Architecture of CCU-MaxNet. U-MaxNet is first designed to extract multimodal hierarchical features from MSRS data by modeling their long-range correlations for cross-modal reconstruction of the source and target domains. Then, MCC-DANet with the MDTNet S-T and T-S models is devised to obtain the generated target domain and reconstructed source domain data under the guidance of the multimodal compensation consistency and adversarial training, eliminating domain shifts at both pixel and semantic levels between different domains. Finally, by integrating them into a unified framework, more discriminative domain-invariant features are carefully extracted for better cross-domain end-to-end classification of MSRS data.

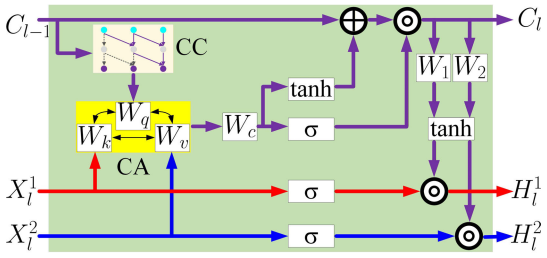


Fig. 2. Inner structure of the xMLSTM cell.

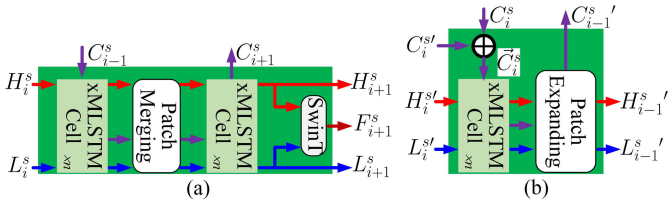


Fig. 3. Inner structure of xMLSTM blocks. (a) Down-block. (b) Up-block.

matrices W_q, W_k, W_v, W_c, W_1, and W_2, respectively. $f_{cc}(\cdot)$ is the causal convolutional (CC) layer with Sigmoid gated linear unit (SiLU) function [55], $f_{ca}(\cdot)$ is the cross-attention function, and $\sigma(\cdot)$ and $tanh(\cdot)$ are the Sigmoid and Tanh functions, respectively. $\circ$ represents the Hadamard product.

For convenience, the above process is defined as $(H_l^1, H_l^2, C_l) = f_x((X_l^1, X_l^2); C_{l-1})$. To adapt it to various tasks, the xMLSTM down-block and up-block are further built as two universal blocks, as shown in Fig. 3. Notably, the memory state provides guidance for exploring more effective interaction mechanisms of multimodal information flows with heterogeneity, which avoids their direct interaction, thus extracting more complementary multimodal features.

2) xMLSTM Down-Block: Fig. 3(a) shows the structure of the xMLSTM down-block, containing some xMLSTM cells, a patch merging operation, and a Swin Transformer (SwinT) [52] block. Taking the source domain and the ith down-block as an example. Under the guidance of memory state C_{i-1}^s, n xMLSTM cells are first used to extract the multimodal features from inputs H_i^s and L_i^s. After the patch merging operation for dimensionality reduction, the other n xMLSTM cells are used to obtain the updated outputs C_{i+1}^s, H_{i+1}^s, and L_{i+1}^s, which are fed into the SwinT block to obtain the fusion features F_{i+1}^s. The calculation process of xMLSTM down-block is written as

$$\begin{aligned}(H_i^1, H_i^2, C_i^s) &= f_{pm}\left(f_x\left(\left(H_i^s, L_i^s\right); C_{i-1}^s\right)\right)\\ (H_{i+1}^s, L_{i+1}^s, C_{i+1}^s) &= f_x\left(\left(H_i^1, H_i^2\right); C_i^s\right)\\ F_{i+1}^s &= f_{st}\left(\left[H_{i+1}^s, L_{i+1}^s\right]\right)\end{aligned} \tag{2}$$

where $f_{pm}(\cdot)$ denotes the patch merging operation, $f_{st}(\cdot)$ is the SwinT block, and $[\cdot,\cdot]$ represents the concatenation operation.

3) xMLSTM Up-Block: Fig. 3(b) shows its inner structure with some xMLSTM cells and patch expanding operation. Take the ith up-block as an example, where $H_i^{s\prime}$ and $L_i^{s\prime}$ are the multimodal inputs, and $C_i^{s\prime}$ is the input memory state from the previous up-block. Unlike traditional MAE-based models [26], [34], [35] or U-shape networks [39], [61], [62], C_i^s from the ith down-block is used instead of H_i^s and L_i^s for cross-modal reconstruction, which can explore the complementarity of MSRS data and reduce the optimization complexity.

From Fig. 3(b), first, $C_i^{s\prime}$ and C_i^s are fused to obtain the enhanced $\vec{C}_i^s$. Then, the n xMLSTM cells are executed, followed by a patch expanding operation to yield the upsampled features $H_{i-1}^{s\prime}$ and $L_{i-1}^{s\prime}$, and the updated memory state

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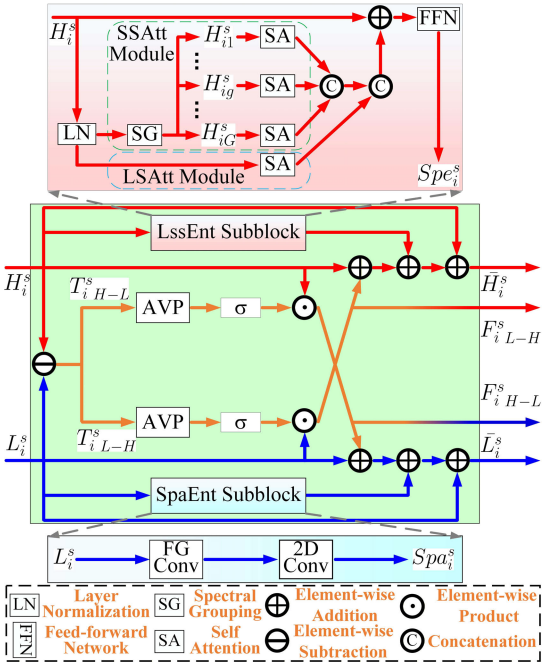


Fig. 4. Inner structure of the CMDC block.

$C_{i-1}^{s}{}'$. The formulas of xMLSTM up-block are described as

$$\vec{C}_i^s = C_i^{s\prime} + C_i^s$$
$$\left(H_{i-1}^{s}{}', L_{i-1}^{s}{}', C_{i-1}^{s}{}'\right) = f_{pe}\left(f_x\left(\left(H_i^{s\prime}, L_i^{s\prime}\right); \vec{C}_i^s\right)\right) \tag{3}$$

where $f_{pe}(\cdot)$ denotes the patch expanding operation.

C. Multimodal Hierarchical Difference Compensation

To explore complementary modality-specific information of MSRS data, a CMDC block is built in Fig. 4, with a long short-term spectral enhancement (LssEnt) subblock and a spatial enhancement (SpaEnt) subblock, where the modality-specific features are extracted from one modality and compensated to another modality.

From Fig. 4, the modality difference learning is first performed on multimodal inputs H_i^s and L_i^s by subtracting their respective elements from each other, yielding cross-modal difference features $T_{i\,H-L}^s$ and $T_{i\,L-H}^s$. They are fused with the original inputs to obtain cross-modal difference enhancement features F_{iH-L}^s and F_{iL-H}^s. Then, the LssEnt subblock $f_{lss}(\cdot)$ and SpaEnt subblock $f_{Spa}(\cdot)$ are used to extract spectral enhancement features Spe_i^s from HSI modality and texture enhancement features Spa_i^s from LiDAR modality, respectively. Finally, by integrating the cross-modal difference enhancement, spectral enhancement, and texture enhancement features and multimodal inputs, the final enhanced multimodal features $\bar{H}_i^s$ and $\bar{L}_i^s$ can be obtained and expressed as

$$\begin{aligned}
T_{i\,H-L}^s &= H_i^s \ominus L_i^s, T_{i\,L-H}^s = L_i^s \ominus H_i^s \\
F_{i\,H-L}^s &= \sigma\left(f_{AVP}\left(T_{i\,H-L}^s\right)\right) \odot H_i^s \\
F_{i\,L-H}^s &= \sigma\left(f_{AVP}\left(T_{i\,L-H}^s\right)\right) \odot L_i^s \\
Spe_i^s &= f_{lss}\left(H_i^s\right), Spa_i^s = f_{Spa}\left(L_i^s\right) \\
\bar{H}_i^s &= H_i^s + F_{i\,L-H}^s + Spe_i^s \\
\bar{L}_i^s &= L_i^s + F_{i\,H-L}^s + Spa_i^s
\end{aligned} \tag{4}$$

where $f_{AVP}(\cdot)$ is the average pooling (AVP) layer, and $\ominus$ and $\odot$ are the element-based subtraction and product, respectively.

For the LssEnt subblock, because there are strong correlations between different bands in HSIs [13], a short-term spectral attention (SSAtt) module and a long-term spectral attention (LSAtt) module are designed to model the short-term and long-term dependencies along the spectral dimension, respectively. In the SSAtt module, after spectral grouping, the HSI features H_i^s are divided into G groups, with the gth group represented as H_{ig}^s, $g = 1, 2, \ldots, G$. We calculate the self-attention operation within each group, whose results are cascaded to obtain the local spectral enhancement features $F_{i\,local}^s$. In the LSAtt module, we directly perform a self-attention operation on H_i^s to capture the global spectral enhancement features $F_{i\,global}^s$. Hence, the spectral enhancement features Spe_i^s are defined as

$$\begin{aligned}
H_{i1}^s, \ldots, H_{iG}^s &= f_{sg}\left(H_i^s\right) \\
F_{i\,local}^s &= \left[f_{sa1}\left(H_{i1}^s\right), \ldots, f_{saG}\left(H_{iG}^s\right)\right] \\
F_{i\,global}^s &= f_{sa}\left(H_i^s\right) \\
Spe_i^s &= FFN\left(\left[F_{i\,local}^s, F_{i\,global}^s\right] + H_i^s\right)
\end{aligned} \tag{5}$$

where $f_{sg}(\cdot)$ is spectral grouping, $f_{sag}(\cdot)$ and $f_{sa}(\cdot)$ are self-attention mechanism, and $FFN(\cdot)$ is feed-forward network [63].

For the SpaEnt subblock, inspired by the fractional Gabor convolutional (Conv) layer designed to describe the multiscale spatial, directional texture, and frequency variation information [15], the fractional Gabor Conv and 2-D Conv layers are utilized in series to extract texture enhancement features Spa_i^s by enhancing modality-specific information of LiDAR data.

D. U-MaxNet for Multimodal Multiscale Feature Extraction

By integrating the above-designed three blocks, an encoder–decoder U-MaxNet is built to fully utilize the multimodal complementarity, extracting representative multimodal multiscale features from MSRS data through cross-modal reconstruction.

For the U-MaxNet encoder in the top row of Fig. 5, it contains stem blocks, cross-scale patch embeddings [64], multimodal window masking [39], CMDC blocks, and xMLSTM down-blocks. $\mathcal{X}_H^s$ and $\mathcal{X}_L^s$ are first fed into the stem blocks to align their dimensions, each of which contains some 2-D Conv blocks with Conv layer, batch normalization (BN) layer, and hard-swish (H-Swish) function, yielding H^s and L^s with channel number c_0. After cross-scale patch embeddings and window masking with masking ratio θ (i.e., ratio of invisible patches), the above features are partitioned into the patch formats, obtaining H_0^s and L_0^s with channel number c_1. The memory state C_0^s is randomly initialized. Then, l CMDC blocks and xMLSTM down-blocks are alternately executed to extract multimodal features H_i^s and L_i^s, updated memory state is C_i^s, and cross-modal difference enhancement features are $F_{(i-1)_{L-H}}^s$ and $F_{(i-1)_{H-L}}^s$ for source MSRS data. Their channel numbers are $c_i = 2^{(i-1)}c_1$, $i = 1, 2, \ldots, l$. Similarly, for the target MSRS data, their patch formats are H_0^t and L_0^t, the extracted features are H_i^t and L_i^t, memory state is C_i^t, and cross-modal difference enhancement features are $F_{(i-1)_{L-H}}^t$ and $F_{(i-1)_{H-L}}^t$.

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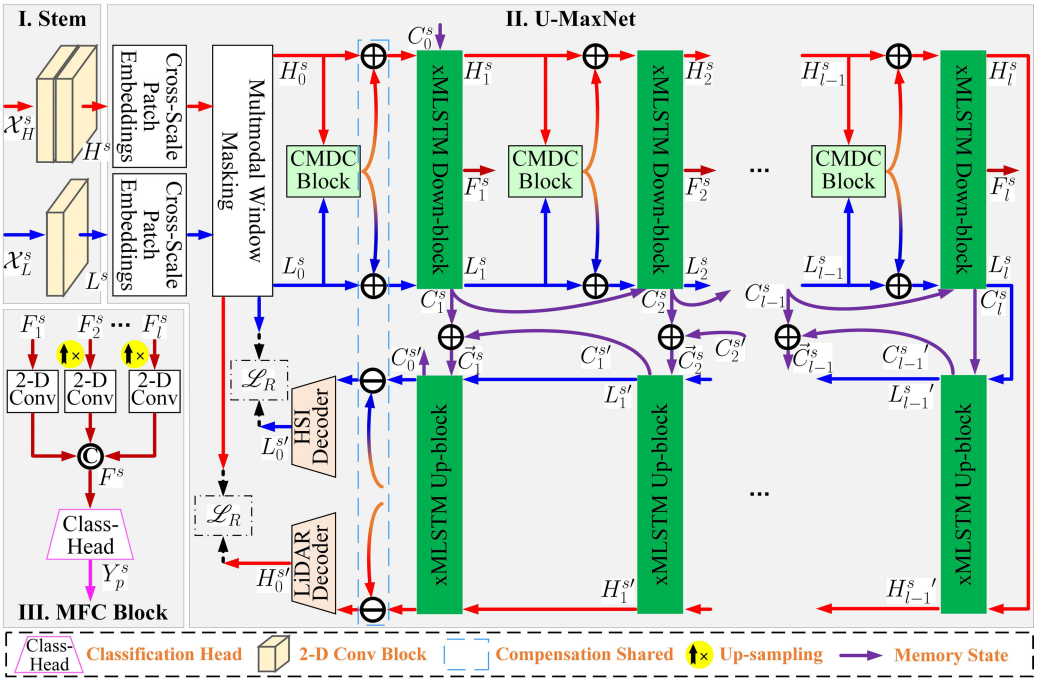


Fig. 5. Structure of the designed U-MaxNet. In Stage I, some stem blocks are used for channel alignment. A U-MaxNet is built for cross-modal reconstruction and feature extraction in Stage II. Stage III applies a MFC block for feature fusion and classification of MSRS data.

Furthermore, the U-MaxNet decoder is devised for cross-modal reconstruction. From the bottom row of Fig. 5, it consists of l xMLSTM down-blocks, an HSI-specific decoder, and a LiDAR-specific decoder. The outputs H_l^s and L_l^s yielded by encoder are first fed into l xMLSTM up-blocks to recover their spatial resolutions layer by layer, obtaining $H_j^{s\prime}$ and $L_j^{s\prime}$, $j = 0, 1, \ldots, l-1$, where the memory state C_{j+1}^s is reused with skip connection to supplement multimodal information layer by layer. Then, by reusing the cross-modal difference enhancement features F_{0L-H}^s and F_{0H-L}^s, the refined features are fed into two specific decoders to reconstruct MSRS data $H_0^{s\prime}$ and $L_0^{s\prime}$ for the source domain data, respectively. The above forward propagation process can be represented as

$$\begin{aligned}
\left(H_j^{s\prime}, L_j^{s\prime}, C_j^{s\prime}\right) &= f_{PEx}\left(f_x\left(H_{j+1}^s, L_{j+1}^s; C_{j+1}^s\right)\right), j = l-1 \\
\vec{C}_{j+1}^s &= C_{j+1}^s + C_{j+1}^{s\prime}, j = 0, 1, \ldots l-2 \\
\left(H_j^{s\prime}, L_j^{s\prime}, C_j^{s\prime}\right) &= f_{PEx}\left(f_x\left(H_{j+1}^{s\prime}, L_{j+1}^{s\prime}; \vec{C}_{j+1}^s\right)\right) \\
H_0^{s\prime} &= f_{hd}\left(H_0^{s\prime} \ominus F_{0L-H}^s\right) \\
L_0^{s\prime} &= f_{ld}\left(L_0^{s\prime} \ominus F_{0H-L}^s\right)
\end{aligned} \tag{6}$$

where $f_{hd}(\cdot)$ and $f_{ld}(\cdot)$ are the HSI- and LiDAR-specific decoders with FC layers, respectively. For the target domain, the corresponding reconstructed MSRS data are $H_0^{t\prime}$ and $L_0^{t\prime}$.

Particularly, according to Fig. 5, different from other deep learning components, the proposed xMLSTM down-block and up-block can assist the designed U-MaxNet in effectively modeling the long-range dependencies in both horizontal and vertical directions under the guidance of the memory state. Particularly, the horizontal dependency learning refers to exploring the complementarity between multimodal features with heterogeneity within each xMLSTM cell, and the vertical dependency learning represents modeling the correlations between multimodal features output by different xMLSTM down-blocks or different xMLSTM up-blocks. Therefore, the above learning mechanism is conducive to the efficient learning and utilization of complementarity between MSRS data and cross-layer features by U-MaxNet to extract more representative multimodal fusion features for downstream tasks.

E. Multimodal Fusion and End-to-End Classification

An multimodal fusion and classification (MFC) block is built for feature fusion. First, the extracted features by the U-MaxNet encoder are fed into an alignment and fusion subblock that is built by using two up-sample layers with bilinear interpolation and three 2-D 1×1 Conv layers with channel number c_f to align their spatial and channel dimensions. By concatenating them along the channel dimension, the multimodal fusion features F^s are obtained. In addition, the fusion features are obtained as F^t for the target domain.

Then, F^s are fed into an end-to-end classification head that is built with an up-sample layer, a 2-D 3×3 Conv block, and a Conv layer to predict the final classification map Y_p^s for the source domain, where the classification loss is defined as

$$\begin{aligned}
Y_p^s &= f_{2D1}^{c_o \to N}\left(f_{2D3}^{3c_f \to c_o}\left(f_{up}\left(F^s\right)\right)\right) \\
\mathscr{L}_{cls}^s &= \mathscr{L}_{CE}\left(Y^s, Y_p^s\right) + \mathscr{L}_R\left(H_0^s, H_0^{s\prime}\right) + \mathscr{L}_R\left(L_0^s, L_0^{s\prime}\right) \\
\mathscr{L}_{mse}^t &= \mathscr{L}_R\left(H_0^t, H_0^{t\prime}\right) + \mathscr{L}_R\left(L_0^t, L_0^{t\prime}\right)
\end{aligned} \tag{7}$$

where $f_{up}(\cdot)$ is the up-sample layer with bilinear interpolation, and $f_{2D3}(\cdot)$ is the 2-D 3×3 Conv block with output channel c_o, which contains the 2-D 2×2 Conv layer, BN layer, and H-Swish function. $f_{2D1}(\cdot)$ is the 2-D 1×1 Conv layer with output channel N, $\mathscr{L}_{CE}(\cdot,\cdot)$ refers to the cross entropy loss, and $\mathscr{L}_R(\cdot,\cdot)$ means the mean squared error (MSE) loss.

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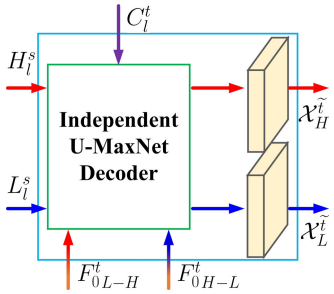


Fig. 6. Structure of the MDTNet S-T.

F. MCC-DANet for Multimodal DA

Inspired by the cycle-consistent for DA [23], a MCC-DANet is designed for domain distribution alignment of MSRS data between different domains. From Fig. 1, it contains a multimodal domain transfer subnetwork (MDTNet) S-T and an MDTNet T-S, obtaining the generated target domain data and reconstructed source domain data, respectively. The reconstructed source domain data are applied to encourage the retention of inherent contents during domain transfer, while the generated target domain data are considered as a co-domain for domain alignment. Then, a multilevel adversarial domain alignment block is constructed, containing a pixel-level alignment subblock and a semantic-level alignment subblock, effectively eliminating domain gaps and improving classification performance of MSRS data in the target domain.

From Fig. 6, MDTNet S-T contains an independent U-MaxNet decoder and two modality generation blocks. The modality-specific features H_l^s and L_l^s in the source domain are first fed into the independent U-MaxNet decoder, while reusing the memory state C_l^t, and the cross-modal difference enhancement features F_{0L-H}^t and F_{0H-L}^t in the target domain for domain transformation. The structure of the independent U-MaxNet decoder is the same as that in Section III-D; however, they are independent of each other. Then, two 2-D Conv blocks are executed to generate target domain data (i.e., $\widetilde{\mathcal{X}_H^t}$ and $\widetilde{\mathcal{X}_L^t}$). Moreover, the reconstruction loss $\mathscr{L}_{mse}^t$ in (7) is used in the above domain transformation process, to ensure that the distribution of the generated target domain data is closer to that of the real target domain data, while optimizing the independent U-MaxNet decoder. It is worth noting that by introducing some independent convolutional layers, the fusion features F^s can also be used as inputs of the above decoder to generate target domain data and achieve good classification results. To make the above process more interpretable, the modality-specific features are utilized here.

Then, for MDTNet T-S, its structure is the same as Fig. 6; however, the independent U-MaxNet decoder is replaced with the decoder trained in the real source domain. First, the generated $\widetilde{\mathcal{X}_H^t}$ and $\widetilde{\mathcal{X}_L^t}$ are fed into U-MaxNet to obtain the modality-specific features $\widetilde{H_l^t}$ and $\widetilde{L_l^t}$, and their fusion representations $\widetilde{F^t}$. Then, the memory state C_i^s and cross-modal difference enhancement features F_{0L-H}^s and F_{0H-L}^s from the real source domain are reused, and combined with the above modality-specific features $\widetilde{H_l^t}$ and $\widetilde{L_l^t}$ to yield the reconstructed source domain data $\widetilde{\mathcal{X}_H^s}$ and $\widetilde{\mathcal{X}_L^s}$. Similar to (7), their prediction result can be obtained and denoted as $\widetilde{Y_p^s}$.

In the above domain transfer process, Y^s and $\widetilde{Y_p^s}$ are used to design an auxiliary classification loss $\mathscr{L}_{cls}^{\widetilde{s}}$, and a cycle-consistent loss $\mathscr{L}_{cyc}$ is defined between the real source domain data and the reconstructed ones for optimizing MCC-DANet

$$\begin{aligned}\mathscr{L}_{cls}^{\widetilde{s}} &= \mathscr{L}_{CE}\left(Y^s, \widetilde{Y_p^s}\right) + \mathscr{L}_{mse}^t \\ \mathscr{L}_{cyc} &= \mathscr{L}_1\left(\mathcal{X}_H^s, \widetilde{\mathcal{X}_H^s}\right) + \mathscr{L}_1\left(\mathcal{X}_L^s, \widetilde{\mathcal{X}_L^s}\right)\end{aligned} \tag{8}$$

where $\mathscr{L}_1(\cdot,\cdot)$ means the $l1$ loss.

Furthermore, in the domain alignment process, the multi-level adversarial domain alignment block is built to reduce the domain gap between the source and target domains. On the one hand, the pixel-level alignment subblock is designed, where two discriminators D_H and D_L are used to classify the generated target domain data from the real ones, achieving domain alignment at the pixel level. The discriminator loss $\mathscr{L}_{pdis}$ is represented as

$$\begin{aligned}\mathscr{L}_{pdis} = &\mathbb{E}_{\mathcal{X}_H^t}\left[\left(D_H\left(\mathcal{X}_H^t\right)-1\right)^2\right] + \mathbb{E}_{\widetilde{\mathcal{X}_H^t}}\left[\left(D_H\left(\widetilde{\mathcal{X}_H^t}\right)-0\right)^2\right] \\ &+ \mathbb{E}_{\mathcal{X}_L^t}\left[\left(D_L\left(\mathcal{X}_L^t\right)-1\right)^2\right] + \mathbb{E}_{\widetilde{\mathcal{X}_L^t}}\left[\left(D_L\left(\widetilde{\mathcal{X}_L^t}\right)-0\right)^2\right]\end{aligned} \tag{9}$$

and the corresponding adversarial loss $\mathscr{L}_{padv}$ is denoted as

$$\begin{aligned}\mathscr{L}_{padv} = &\mathbb{E}_{\widetilde{\mathcal{X}_H^t}}\left[\left(D_H\left(\widetilde{\mathcal{X}_H^t}\right)-1\right)^2\right] \\ &+ \mathbb{E}_{\widetilde{\mathcal{X}_L^t}}\left[\left(D_L\left(\widetilde{\mathcal{X}_L^t}\right)-1\right)^2\right].\end{aligned} \tag{10}$$

On the other hand, the semantic-level alignment subblock applies another discriminator D_F to classify the fusion features extracted from the generated target domain data and those from the real ones, in order to perform domain alignment at the semantic level. The discriminator loss $\mathscr{L}_{sdis}$ is written as

$$\mathscr{L}_{sdis} = \mathbb{E}_{F^t}\left[\left(D_F\left(F^t\right)-1\right)^2\right] + \mathbb{E}_{\widetilde{F^t}}\left[\left(D_F\left(\widetilde{F^t}\right)-0\right)^2\right] \tag{11}$$

and the corresponding adversarial loss $\mathscr{L}_{sadv}$ is expressed as

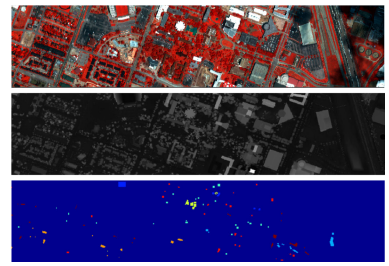
$$\mathscr{L}_{sadv} = \mathbb{E}_{\widetilde{F^t}}\left[\left(D_F\left(\widetilde{F^t}\right)-1\right)^2\right]. \tag{12}$$

By integrating the above designs into a unified CCU-MaxNet framework, the total classification loss $\mathscr{L}_{cls}$, discriminator loss $\mathscr{L}_{dis}$, and adversarial loss $\mathscr{L}_{adv}$ are defined as

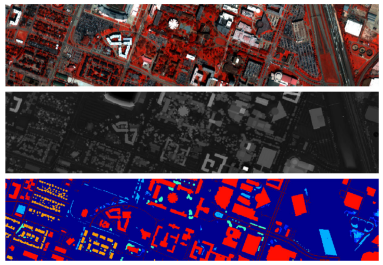
$$\begin{aligned}\mathscr{L}_{cls} &= \mathscr{L}_{cls}^s + \mathscr{L}_{cls}^{\widetilde{s}} + \mathscr{L}_{cyc} \\ \mathscr{L}_{dis} &= \mathscr{L}_{pdis} + \mathscr{L}_{sdis} \\ \mathscr{L}_{adv} &= \mathscr{L}_{padv} + \mathscr{L}_{sadv}.\end{aligned} \tag{13}$$

It should be noted that different from the existing methods [1], [7], [28], [29], [30], [31], [33], [39], the designed memory state not only serves as the guidance for enhancing multimodal feature extraction in the U-MaxNet but also assists MCC-DANet in learning more discriminative domain-invariant representations, thus effectively improving the cross-domain MSRS data classification of the proposed CCU-MaxNet framework.

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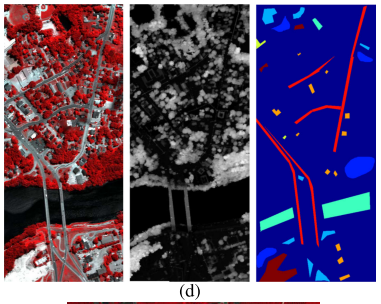
(a)



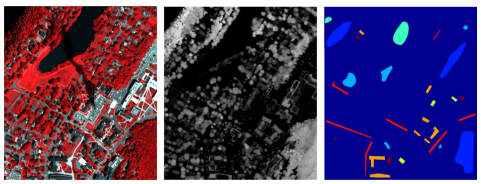
(b)

Class			Number of Samples	
NO.	Color	Name	Source	Target
1		Grass healthy	345	1353
2		Grass stressed	365	4888
3		Trees	365	2007
4		Water	285	22
5		Commercial	319	5347
6		Non-commercial	408	32459
7		Roads	443	6365
Total			2530	52441

(c)



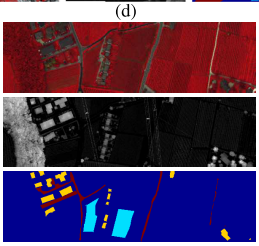
(d)



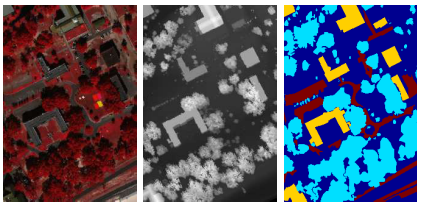
(e)

Class			Number of Samples	
NO.	Color	Name	Source	Target
1		Trees	12713	18605
2		Grass	7528	4410
3		Water	16323	5425
4		White roof	698	860
5		Grey roof	3417	4338
6		Roads	17476	4955
7		Bare soil	8821	764
Total			67076	38457

(f)



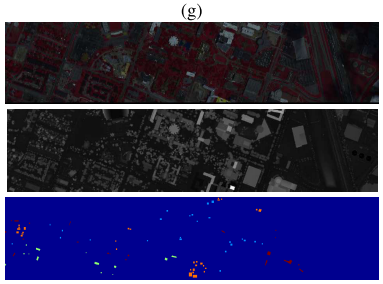
(g)



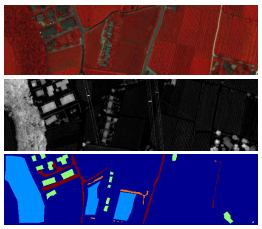
(h)

Class			Number of Samples	
NO.	Color	Name	Source	Target
1		Trees	4034	23246
2		Buildings	2903	6240
3		Roads	3174	6687
Total			10111	36173

(i)



(j)



(k)

Class			Number of Samples	
NO.	Color	Name	Source	Target
1		Grounds	365	9123
2		Buildings	319	2903
3		Trees	650	479
4		Roads	443	3174
Total			1777	15679

(l)

Fig. 7. (Top or Left) False-color map of HSI data, (middle) gray-scale map of LiDAR data, and (bottom or right) ground-truth map. (a)–(c) H13–H18 dataset (HSI; LiDAR; number of source and target samples). (d)–(f) NA–HA dataset (HSI; LiDAR; number of source and target samples). (g)–(i) TO–MU dataset (HSI; LiDAR; number of source and target samples). (j)–(l) H13–TO dataset (HSI; LiDAR; number of source and target samples).

IV. Experimental Results

Four cross-domain MSRS datasets, including Houston13–Houston18 (H13–H18), Nashua–Hanover (Na–Ha), Trento-Muufl (TO–MU), and Houston13–Trento (H13–TO), are used to measure the performance of our CCU-MaxNet. Overall accuracy (OA), average accuracy (AA), and Kappa (κ) coefficient are used as the quantitative metrics, whose average values after five runs are reported. All experiments are conducted using an Nvidia GeForce RTX 4070 GPU and a 3.50 GHz Intel Core i5-13600KF processor with PyTorch 2.0 and Python 3.10.

A. Experimental Data

1) H13–H18 Dataset: There are two scenes with co-registered HSI and LiDAR data: Houston13 (source) and Houston18 (target). They were collected over the University of Houston campus and its neighbor area, Houston, TX, USA, and released by the IEEE Geoscience and Remote Sensing Society Data Fusion Contest in 2013[1] and 2018,[2] respectively. Their overlap area contains 210 $\times$ 954 pixels, seven land covers, and 48 bands for cross-temporal analysis, as shown in Fig. 7(a)–(c).

2) Na–Ha Dataset: This dataset contains Nashua (source) and Hanover (target) data, which were respectively collected in 2014 over Nashua and Hanover, USA, whose raw HSI and LiDAR data come from G-LiHT data.[3] There are 450 $\times$ 1050 and 720 $\times$ 620 pixels with ground sample distance (GSD) 1 m in the Nashua and Hanover data, respectively, with 114 spectral bands ranging from 0.42 to 0.95 μm and seven land covers. The false-color map, gray-scale map, ground-truth map, and corresponding sample number are shown in Fig. 7(d)–(f).

[1][Online]. Available: http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

[2][Online]. Available: http://www.grss-ieee.org/community/technical-committees/data-fusion/2018-ieee-grss-data-fusion-contest/

[3][Online]. Available: https://glihtdata.gsfc.nasa.gov

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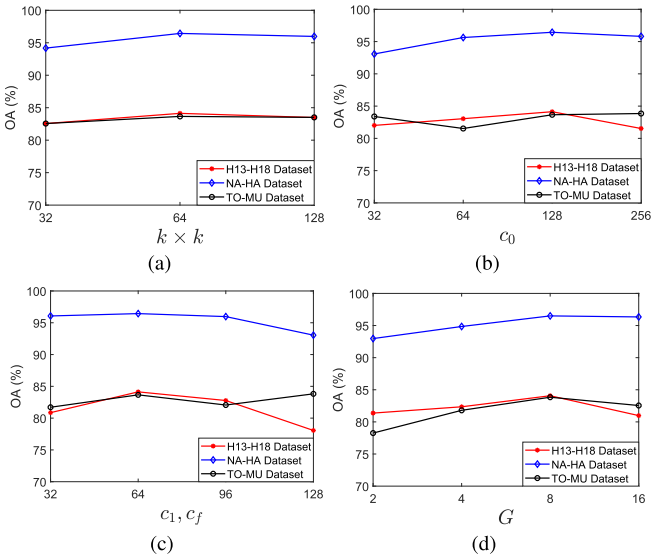


Fig. 8. Parameter analysis. (a) $k \times k$. (b) c_0. (c) c_1 and c_f. (d) G.

3) TO-MU Dataset: Based on [1], the Trento and Muufl data are set as the source and target domains, respectively. Trento data were collected over the rural area of Trento, Italy, which contains 600 × 166 pixels with a spatial resolution of 1 m, and 63 spectral bands in the wavelength range of 420.89 to 989.09 nm. Muufl data were captured over the University of Southern Mississippi's Gulf Park Campus, Long Beach, Mississippi, USA,[4] where there are 325 × 220 pixels with a spatial resolution of 1 m, and 64 spectral bands in the wavelength range of 375–1050 nm. In addition, three land covers are selected for analysis, as reported in Fig. 7(g)–(i).

4) H13–TO Dataset: From [28], Houston13 and Trento were used as the source and target domains, respectively. 63 bands and four classes are extracted for cross-area analysis.

Inspired by [33] and [41], and with the same settings as [39], the patch-free method is adopted in the proposed CCU-MaxNet for cross-domain classification of MSRS data.

B. Parameter Setting Analysis

The first experiment is designed for parameter analysis, containing image size k, channel number c_0 in the stem blocks, c_1 after patch embeddings, c_f in the MFC block, and spectral group number G. Their candidate sets are defined as $k \in \{32, 64, 128\}$, $c_0 \in \{32, 64, 128, 256\}$, $c_1, c_f \in \{32, 64, 96, 256\}$, and $G \in \{2, 4, 8, 16\}$, respectively. Fig. 8 reports the corresponding results with the OA metric to analyze the influences of these parameters on the three MSRS datasets.

1) Analysis of k: From Section III-A, k denotes the input size, whose effects on classification accuracy are shown in Fig. 8(a). As its value increases, the OA value presents a trend of first increasing and then decreasing. Our CCU-MaxNet reaches the quasi-optimal accuracy when k is set to 64.

2) Analysis of Number c_0: c_0 is the channel number of stem blocks. As shown in Fig. 8(b), for the H13–H18 and NA–HA datasets, the OA value is the highest when c_0 is 128. For the TO–MU dataset, although the OA value continues to rise when c_0 is greater than 128, for the sake of a unified structure and practical application, c_0 is also set to 128.

3) Analysis of Number c_1 and c_f: The values of c_1 and c_f are related to the complexity of CCU-MaxNet. The smaller their values, the less complex it is, but the more information will be lost. To reduce the complexity of parameter analysis, they are set to be the same. From Fig. 8(c), the OA value reaches a quasi-optimal value when they are fixed as 64.

4) Analysis of Group G: G is the number of spectral groups in (5), whose influences on OA value are illustrated in Fig. 8(d). It can be inferred that the OA value of the whole CCU-MaxNet reaches quasi-optimal when it is set to 8.

In addition, θ is fixed to 75%. l is set to 3 for balancing the accuracy and complexity. The batch size is set to 4, and the initial learning rates for U-MaxNet and MCC-DANet are fixed to 0.0005 with the polynomial learning rate schedule. To facilitate practical application and further prove the transferability of the proposed framework, the above parameters are fixed and directly applied to the H13–TO dataset.

C. Classification Performance Analysis

The second experiment is constructed for classification performance analysis between the proposed CCU-MaxNet framework and some competitive models, containing SCLUDA [25], MLUDA [27], MSDA [26], FrF-SSDA [28], MDA-NET [29], HighDAN [33], SASS [31], and HMDA-DANet [39]. Their classification results (i.e., class-specific accuracy, OA, AA, and κ) for the four cross-domain MSRS datasets are reported in Tables I–IV, followed by the visualization maps in Figs. 9–12.

1) Structure Analysis: SCLUDA, MSDA, and MLUDA were originally designed for cross-domain HSI classification. Due to the lack of multimodal designs, HSI and LiDAR data are directly cascaded at the pixel level, whose strong heterogeneity limits the cross-domain performance of the above methods, even lower than their performance using only HSI. FrF-SSDA built a fractional differential mask and a fractional fusion module for fusing data and reducing the spectral shift of HSIs. In MDA-NET, the optimal transport constrained dynamic normalization and heterogeneous knowledge graphs matching methods were designed for domain alignment of topological structures. SASS introduced the shared-private feature, decoupling and class-balanced pseudo-label generation modules for cross-domain learning. However, the above methods are all patch-based, and the patch-by-patch processing approach leads to a lack of effective utilization of global information, which in turn limits their classification accuracy. HighDAN and HMDA-DANet were two patch-free algorithms. HighDAN introduced the high-resolution network to extract high-to-low-resolution features, followed by adversarial DA for cross-city segmentation. HMDA-DANet was one of our previous works, which designs the first multimodal MAE without pretraining, followed by a soft domain encoding-based discriminator and an entropy-based discriminator for semantic feature and prediction structure alignment. Although these methods achieved good classification results, they ignore the utilization of cross-layer dependencies and differential information of MSRS data, which may make them prone to

[4][Online]. Available: https://github.com/GatorSense/MUUFLGulfport

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TABLE I

CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS FOR THE H13–H18 DATASET

Class	SCLUDA	MSDA	MLUDA	FrF-SSDA	MDA-NET	HighDAN	SASS	HMDA-DANet	Proposed (w/o FG)	Proposed (w/ FG)
1	85.89 ± 11.52	**97.22 ± 1.26**	68.36 ± 19.55	73.60 ± 24.91	79.01 ± 11.49	71.43 ± 12.12	61.31 ± 9.91	83.10 ± 4.96	83.86 ± 4.41	72.27 ± 14.27
2	56.24 ± 11.74	41.67 ± 4.29	75.02 ± 14.22	57.22 ± 14.25	74.85 ± 6.54	62.90 ± 6.16	**86.30 ± 3.87**	55.67 ± 5.49	71.85 ± 5.49	81.93 ± 6.41
3	83.81 ± 6.82	85.86 ± 6.77	85.32 ± 5.43	86.75 ± 6.47	86.92 ± 2.65	94.96 ± 1.83	91.21 ± 4.78	94.50 ± 1.66	**95.35 ± 2.63**	90.66 ± 4.15
4	77.27 ± 27.27	85.45 ± 19.92	**100.00 ± 0.00**	92.73 ± 11.85	92.10 ± 11.26	76.36 ± 5.93	95.45 ± 6.43	**100.00 ± 0.00**	81.82 ± 0.00	89.09 ± 9.96
5	88.06 ± 3.01	**91.47 ± 6.10**	82.15 ± 8.73	77.90 ± 7.98	40.03 ± 2.28	68.53 ± 5.44	81.00 ± 15.24	76.03 ± 7.88	80.86 ± 5.33	84.52 ± 3.76
6	65.49 ± 2.12	78.29 ± 2.55	77.25 ± 2.87	81.04 ± 3.32	**92.99 ± 4.88**	85.15 ± 2.65	74.84 ± 2.47	87.32 ± 1.62	84.87 ± 2.32	85.53 ± 2.30
7	82.02 ± 4.07	84.34 ± 3.84	82.94 ± 2.39	77.09 ± 5.58	**89.58 ± 5.66**	75.05 ± 4.05	79.30 ± 5.84	79.05 ± 6.74	78.50 ± 3.63	78.34 ± 3.97
OA	70.14 ± 1.09	77.70 ± 1.30	78.29 ± 1.77	78.05 ± 0.52	78.88 ± 0.81	80.17 ± 1.96	77.32 ± 1.04	82.38 ± 0.33	82.85 ± 1.37	**84.07 ± 1.46**
AA	76.97 ± 3.22	80.61 ± 3.52	81.58 ± 3.04	78.05 ± 5.68	79.36 ± 2.97	76.34 ± 2.55	81.34 ± 0.81	82.24 ± 0.73	82.44 ± 0.93	**83.19 ± 1.33**
$\kappa \times 100$	56.95 ± 1.23	65.82 ± 1.67	66.84 ± 2.49	65.03 ± 0.44	66.13 ± 2.33	68.49 ± 2.61	65.82 ± 1.29	71.43 ± 0.59	72.64 ± 1.93	**74.47 ± 2.00**

TABLE II

CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS FOR THE NA–HA DATASET

Class	SCLUDA	MSDA	MLUDA	FrF-SSDA	MDA-NET	HighDAN	SASS	HMDA-DANet	Proposed (w/o FG)	Proposed (w/ FG)
1	93.88 ± 2.21	94.82 ± 1.36	90.68 ± 1.23	95.35 ± 0.11	**99.94 ± 0.07**	96.77 ± 1.78	94.56 ± 1.32	94.51 ± 0.62	98.56 ± 0.65	98.80 ± 0.49
2	94.98 ± 10.88	92.93 ± 5.39	**100.00 ± 0.00**	82.93 ± 6.06	97.91 ± 1.20	99.97 ± 0.02	**100.00 ± 0.00**	100.00 ± 0.01	99.94 ± 0.07	**100.00 ± 0.00**
3	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	83.59 ± 9.01	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
4	**100.00 ± 0.00**	99.26 ± 0.25	99.63 ± 0.17	96.47 ± 0.06	93.27 ± 9.33	84.84 ± 15.45	**100.00 ± 0.00**	94.70 ± 5.09	98.56 ± 0.73	98.23 ± 1.34
5	54.39 ± 14.37	**94.39 ± 1.94**	91.11 ± 4.44	80.85 ± 3.50	85.80 ± 7.75	76.15 ± 9.59	87.24 ± 2.96	88.95 ± 4.75	76.14 ± 4.82	85.41 ± 2.82
6	91.15 ± 2.24	88.03 ± 1.27	89.51 ± 1.99	90.85 ± 1.97	82.70 ± 2.27	94.17 ± 7.08	97.32 ± 2.06	**98.72 ± 0.30**	95.65 ± 1.74	92.67 ± 3.09
7	97.83 ± 1.93	23.58 ± 4.28	63.00 ± 22.24	37.38 ± 30.90	**100.00 ± 0.00**	81.49 ± 13.41	50.08 ± 27.98	75.65 ± 11.13	86.05 ± 9.24	85.55 ± 11.82
OA	90.00 ± 1.60	92.91 ± 0.44	92.33 ± 0.64	91.13 ± 0.52	93.25 ± 2.02	94.28 ± 0.70	94.53 ± 0.63	95.33 ± 0.83	95.74 ± 0.75	**96.50 ± 0.65**
AA	90.32 ± 1.16	84.71 ± 0.89	90.56 ± 2.91	83.40 ± 4.32	91.89 ± 2.11	90.48 ± 0.54	89.89 ± 3.35	93.22 ± 1.86	93.56 ± 0.47	**94.38 ± 1.83**
$\kappa \times 100$	86.30 ± 2.18	90.14 ± 0.62	89.50 ± 0.86	87.54 ± 0.73	90.45 ± 2.85	92.00 ± 0.98	92.46 ± 0.86	93.50 ± 1.14	94.00 ± 1.04	**95.08 ± 0.90**

TABLE III

CLASSIFICATION ACCURACY (%) USING DIFFERENT METHODS FOR THE TO–MU DATASET

Class	SCLUDA	MSDA	MLUDA	FrF-SSDA	MDA-NET	HighDAN	SASS	HMDA-DANet	Proposed (w/o FG)	Proposed (w/ FG)
1	83.50 ± 3.78	86.05 ± 4.06	77.61 ± 3.15	72.19 ± 7.72	**98.65 ± 0.77**	85.89 ± 3.81	90.51 ± 9.17	88.17 ± 2.79	75.69 ± 2.57	78.84 ± 5.49
2	42.88 ± 5.86	46.61 ± 10.21	50.42 ± 10.06	55.56 ± 8.58	53.72 ± 18.64	91.45 ± 10.04	30.36 ± 18.68	69.50 ± 5.50	**97.29 ± 2.33**	94.85 ± 4.24
3	57.35 ± 17.02	70.84 ± 9.16	87.83 ± 6.01	87.61 ± 9.92	48.37 ± 7.81	41.67 ± 6.14	**91.78 ± 5.66**	67.56 ± 3.61	90.42 ± 5.07	90.99 ± 4.26
OA	71.63 ± 1.75	76.41 ± 3.41	74.78 ± 2.95	72.18 ± 2.34	80.05 ± 0.65	78.67 ± 2.07	80.19 ± 3.52	81.56 ± 2.09	82.14 ± 1.74	**83.84 ± 2.91**
AA	61.24 ± 5.67	67.83 ± 6.16	71.95 ± 3.52	71.79 ± 2.46	64.30 ± 2.72	73.00 ± 3.88	70.88 ± 3.84	75.08 ± 2.65	87.80 ± 1.54	**88.22 ± 1.05**
$\kappa \times 100$	46.09 ± 5.42	54.21 ± 7.83	56.14 ± 4.65	52.73 ± 1.76	62.38 ± 1.39	61.47 ± 3.48	63.61 ± 4.61	64.23 ± 4.03	69.87 ± 2.71	**72.31 ± 4.12**

TABLE IV

CLASSIFICATION ACCURACY (%) USING DIFFERENT UDA METHODS FOR THE H13–TO DATASET

Class	SCLUDA	MSDA	MLUDA	FrF-SSDA	MDA-NET	HighDAN	SASS	HMDA-DANet	Proposed (w/o FG)	Proposed (w/ FG)
1	**99.95 ± 0.09**	99.84 ± 0.35	99.14 ± 0.67	99.78 ± 0.36	98.41 ± 0.72	98.07 ± 1.17	99.02 ± 0.61	98.45 ± 1.01	98.94 ± 0.95	99.39 ± 0.95
2	57.78 ± 10.40	50.02 ± 36.36	78.18 ± 4.52	91.97 ± 3.82	72.18 ± 12.84	90.32 ± 5.87	93.84 ± 2.88	93.97 ± 5.66	**98.40 ± 1.53**	97.23 ± 1.65
3	19.54 ± 25.71	71.44 ± 40.40	39.42 ± 22.72	85.72 ± 17.46	74.88 ± 17.77	40.42 ± 23.49	74.49 ± 9.59	86.05 ± 5.35	88.60 ± 6.97	**89.02 ± 7.56**
4	84.58 ± 4.23	88.17 ± 8.30	83.03 ± 10.65	80.15 ± 4.62	88.94 ± 9.46	83.22 ± 8.49	84.96 ± 4.70	**91.41 ± 4.11**	88.19 ± 3.54	90.97 ± 3.71
OA	89.30 ± 0.89	89.93 ± 3.21	92.00 ± 1.68	95.13 ± 0.30	91.01 ± 3.03	93.13 ± 0.47	95.39 ± 0.74	96.34 ± 0.79	96.88 ± 0.96	**97.46 ± 0.93**
AA	65.46 ± 5.61	77.37 ± 6.44	74.94 ± 5.75	89.41 ± 3.89	83.60 ± 6.72	78.01 ± 4.53	88.08 ± 2.82	92.44 ± 1.15	93.53 ± 1.08	**94.15 ± 1.13**
$\kappa \times 100$	78.61 ± 1.75	79.89 ± 6.31	84.21 ± 3.38	90.36 ± 0.60	82.12 ± 5.96	86.40 ± 0.79	90.96 ± 1.46	92.84 ± 1.51	93.88 ± 1.82	**95.01 ± 1.76**

some cross-domain classification issues, such as imbalanced classification.

Unlike other methods, our CCU-MaxNet achieves the optimal cross-domain performance in the following three aspects. First, xMLSTM is designed as a new multimodal component, providing a method for exploring long-range correlations between multitemporal multimodal data or multimodal hierarchical features. Second, under the guidance of memory state and multimodal complementarity, U-MaxNet is built to model long-range correlations between modalities and between layers to extract representative fusion features. Finally, with the multimodal differential compensation and cycle-consistent, MCC-DANet is built for domain alignments at both pixel and semantic levels. Particularly, to meet the application requirements of different scenarios, CCU-MaxNet has two variant models with different working modes. The high-precision mode, namely the proposed CCU-MaxNet (w/ FG), enhances its learning ability of directional texture and frequency information by introducing FrFT in the CMDC block to obtain an average gain of about 1.07% in OA value, which is suitable for some scenarios that require high accuracy. In contrast, for some efficiency-focused scenarios, removing FrFT from the entire framework forms its high-efficiency mode, namely the proposed CCU-MaxNet (w/o FG), which almost doubles the inference speed while maintaining classification accuracy better than all comparison methods.

2) Results and Analysis on the H13–H18 Dataset: Table I reports the classification results of all methods in the H13–H18 dataset, whose classification maps are visualized in Fig. 9. Although SCLUDA, MSDA, MLUDA, and FrF-SSDA are designed with various DA methods, they achieve relatively poor performance. For example, MSRS data are directly cascaded along with their channel dimension as the input of SCLUDA, with OA value of only 70.14% and κ value of only 56.95%, which are far lower than its performance when using HSI alone. Through a multimodal reconstruction strategy, MDA-NET and SASS obtain relatively better performance. As a cross-domain segmentation method, HighDAN improves the OA value to 80.17%; however, the simple multimodal fusion and adversarial strategies affect the further improvement of its accuracy, with an AA value of only 76.34%. Thanks to the self-supervised learning, HMDA-DANet improves the

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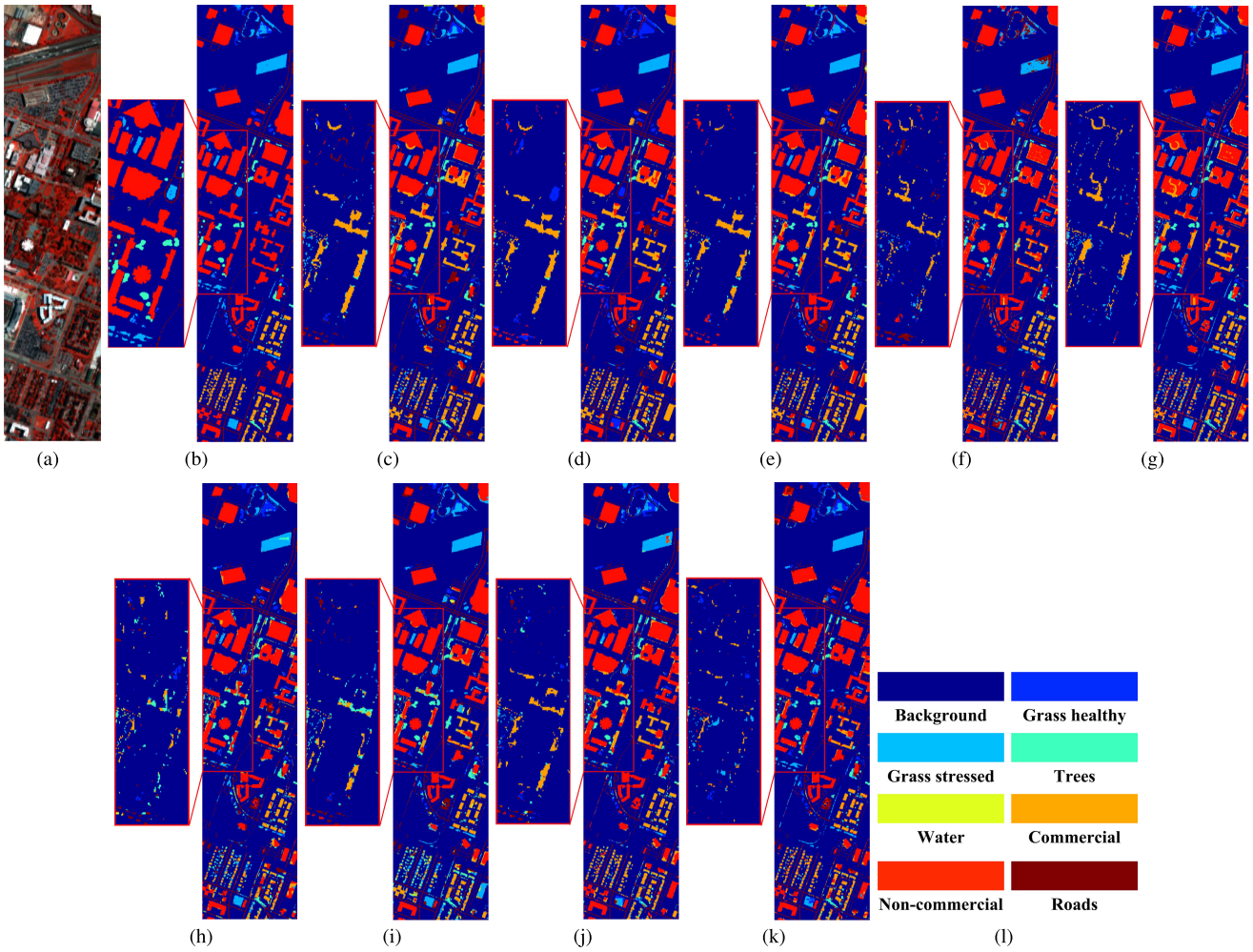


Fig. 9. Classification maps and error maps of some highlighted areas for the H13–H18 dataset. (a) Target HSI (Rband:35, Gband:15, and Bband:10). (b) Target ground-truth map. (c) SCLUDA. (d) MSDA. (e) MLUDA. (f) FrF-SSDA. (g) MDA-NET. (h) HighDAN. (i) SASS. (j) HMDA-DANet. (k) Proposed. (l) Legend.

OA, AA, and κ metrics to 82.38%, 82.24%, and 71.43%, all ranking second. However, it adopts a SwinT-based dual-branch structure and ignores the long-term corrections between modalities and between layers, failing to fully explore the inherent differential representation advantages of MSRS data. Specifically, CCU-MaxNet achieves at least 1.69%, 0.95%, and 3.04% gains in OA, AA, and κ values, respectively. From Fig. 9, although it does not perform optimally in any category, its accuracy for each category exceeds 72.27%, making its classification map closer to the ground-truth map, verifying its effectiveness.

3) Results and Analysis on the NA-HA Dataset: Table II and Fig. 10 compare CCU-MaxNet with respect to other methods on the NA-HA dataset. With the multimodal designs, MDA-NET, HighNAN, and SASS yield the third best OA value, ranging from 93.25% to 94.53%. However, their inaccurate classification of certain land covers results in low AA values, such as HighDAN misclassifying classes "Gray roof" and "Roads" in Fig. 10(h); while MDA-NET and SASS misclassifying classes "Bare soil" and "Roads" in Fig. 10(g) and (i). The self-supervised learning strategy enables the three metrics of HMDA-DANet to exceed 93.22%, achieving the second-best performance, with obvious gains in classification accuracy for most classes, as shown in Fig. 10(j). CCU-MaxNet is strong on nouns and obtains OA value of 96.50%, AA value of 94.38%, and κ value of 95.08%, improving over prior optimal art by 1.17%, 1.16%, and 1.58%, respectively. Hence, its classification map shows fewer misclassified areas in Fig. 10(k). The experimental results further reveal that effective multimodal feature extraction and DA strategies provide guarantees for better performance of CCU-MaxNet.

4) Results and Analysis on the TO–MU Dataset: In Table III, although the TO–MU dataset only contains three classes, large differences in sample distribution and high interclass similarity increase its classification difficulty, where the OA value of most comparison methods is below 78.67%, with only MDA-NET, SASS, and HMDA-DANet exceeding 80.05%. In addition, MSDA, FrF-SSDA, and SASS have some misclassification between classes "Buildings" and "Trees" in Fig. 11(d), (f), and (i), while SCLUDA, MLUDA, MDA-NET, and HighDAN exhibit misclassification between classes "Buildings" and "Roads" in Fig. 11(c), (e), (g), and (h) resulting in low AA values. HMDA-DANet improves the classification performance, but still suffers from poor interclass separability in Fig. 11(j). Due to the insufficient utilization of the imaging advantages of each modality, there are many

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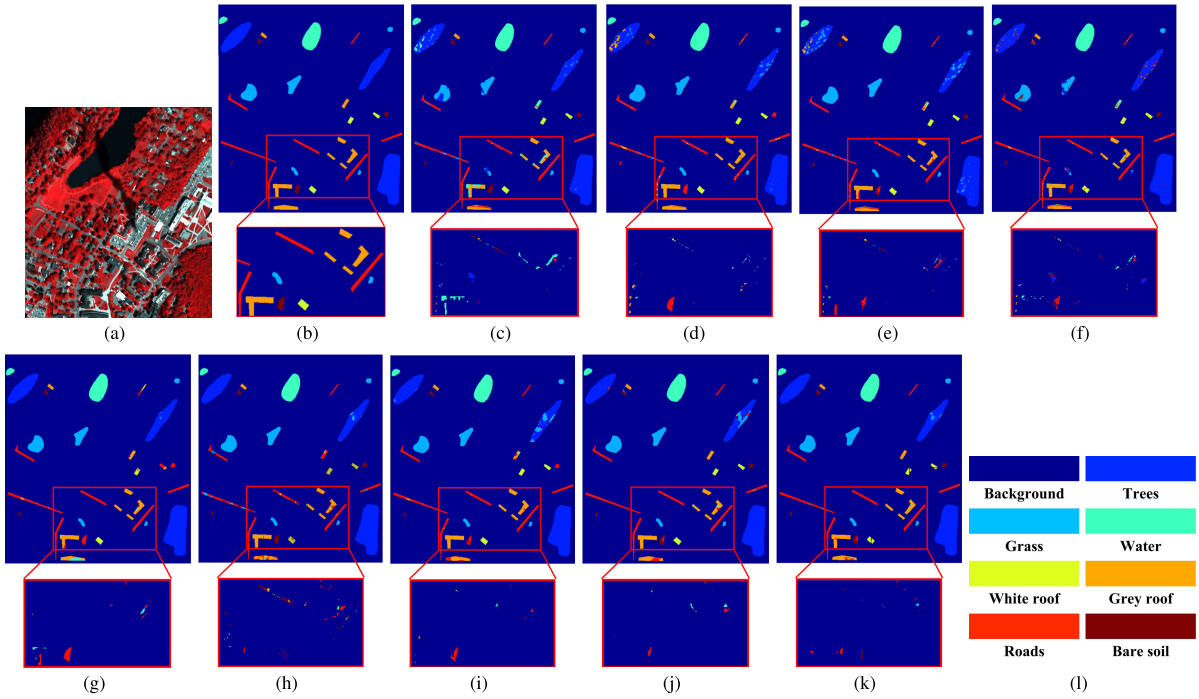


Fig. 10. Classification maps and error maps of some highlighted areas for the NA–HA dataset. (a) Target HSI (Rband:20, Gband:35, and Bband:15). (b) Target ground-truth map. (c) SCLUDA. (d) MSDA. (e) MLUDA. (f) FrF-SSDA. (g) MDA-NET. (h) HighDAN. (i) SASS. (j) HMDA-DANet. (k) Proposed. (l) Legend.

TABLE V

ABLATION STUDY OF THE PROPOSED CCU-MAXNET FRAMEWORK ON THE THREE CROSS-DOMAIN MSRS DATASETS

	Components					H13-H18 Dataset		NA-HA Dataset		TO-MU Dataset	
	Skip Connection	CMDC	xMLSTM	SwinT	MCC-DANet	OA	$\kappa\times100$	OA	$\kappa\times100$	OA	$\kappa\times100$
(a)	✘	✘	✘	✔	✘	72.70	52.65	83.33	76.62	66.04	43.10
(b)	✘	✘	✘	✔	✔	78.92	64.94	88.86	84.36	75.27	55.93
	✘	✘	✔	✘	✔	81.77	69.64	92.05	88.77	79.15	63.71
	✘	✔	✔	✘	✔	83.45	72.76	95.04	93.01	83.10	70.72
(c)	✔	✔	✔	✘	✘	13.35	—	25.13	—	33.61	—
	✔	✘	✔	✘	✔	82.61	71.75	93.43	90.71	82.76	69.15
(d)	✔	✔	✔	✘	✔	**84.07**	**74.47**	**96.50**	**95.08**	**83.84**	**72.31**

misclassified areas and blurred boundaries in the classification maps of the above methods. On the contrary, CCU-MaxNet achieves optimal OA, AA, and κ values, increasing by 2.28%, 13.14%, and 8.08% compared with their optimal values, respectively. From Fig. 11(k), it achieves more accurate boundaries between classes "Buildings" and "Roads," although sacrificing some accuracy for class "Trees." The experimental results also confirm that CCU-MaxNet outperforms other competitors in the cross-domain classification task of MSRS data.

5) Results and Analysis on the H13–TO Dataset: Table IV shows the classification results in the H13–TO dataset, with the classification maps visualized in Fig. 12. Compared with other comparison methods, CCU-MaxNet obtains gains of at least 1.12%, 1.71%, and 2.17% on OA, AA, and κ values, respectively. In addition, from Fig. 12, it achieves the best performance in class "Buildings" and class "Trees," consistent with the results in Table IV. The above results further verify the superiority of the proposed CCU-MaxNet framework.

D. Ablation Study

In Figs. 1 and 5, xMLSTM, CMDC block, memory state, and MCC-DANet are designed for multimodal learning, cross-modal compensation, and domain-invariant representation. The third experiment conducted a detailed ablation study on the first three datasets for assessing their roles using OA and κ values. The Baseline is built with the SwinT block [52] and a discriminator [22], as shown in part (a) of Table V. The proposed framework with and without each component is denoted as proposed (w/) and proposed (w/o), respectively.

1) Analysis of MCC-DANet: MCC-DANet is designed to eliminate cross-domain shifts of MSRS data. We compare this design with two variants, i.g., Baseline+MCC-DANet and the proposed (w/o MCC-DANet). As shown in the first line of part (b) in Table V, with MCC-DANet, the OA value of Baseline is improved from 72.70% to 78.92%, 83.33% to 88.86%, and 66.04% to 75.27% in the three datasets, respectively. Moreover, as shown in part (d) and the first line of part (c) in Table V, MCC-DANet improves the OA value of CCU-MaxNet by 70.72%, 71.37%, and 50.23%, respectively. The experimental results prove the effectiveness of MCC-DANet.

2) Analysis of CMDC Block: To better explore the complementary information of MSRS data and assist in eliminating domain shift, the CMDC block is built. From the last line of parts (b) and (c) in Table V, it brings at least 1.68% and 1.08%

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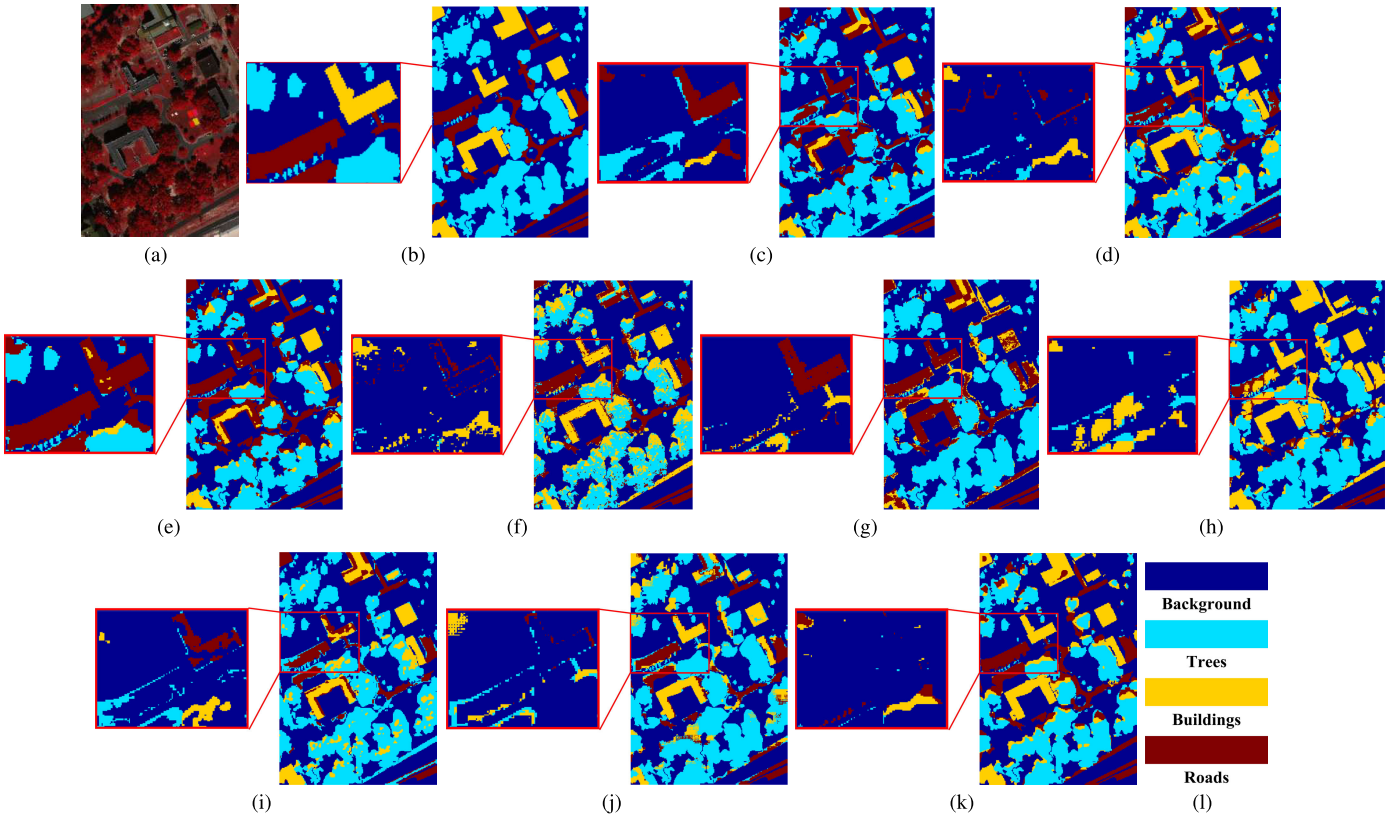


Fig. 11. Classification maps and error maps of some highlighted areas for the TO–MU dataset. (a) Target HSI (Rband:45, Gband:25, and Bband:15). (b) Target ground-truth map. (c) SCLUDA. (d) MSDA. (e) MLUDA. (f) FrF-SSDA. (g) MDA-NET. (h) HighDAN. (i) SASS. (j) HMDA-DANet. (k) Proposed. (l) Legend.

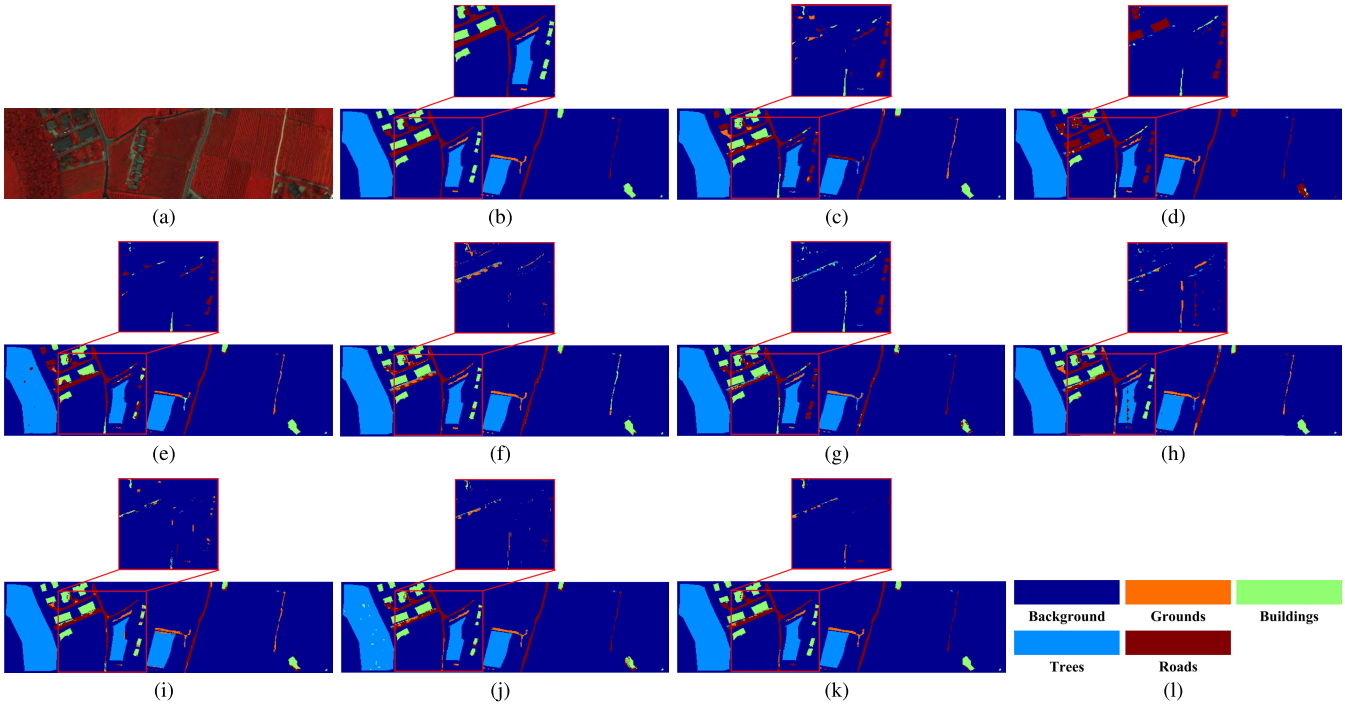


Fig. 12. Classification maps and error maps of some highlighted areas for the H13–TO dataset. (a) Target HSI (Rband:42, Gband:20, and Bband:5). (b) Target ground-truth map. (c) SCLUDA. (d) MSDA. (e) MLUDA. (f) FrF-SSDA. (g) MDANet. (h) HighDAN. (i) SASS. (j) HMDA-DANet. (k) Proposed. (l) Legend.

improvements in OA value to the xMLSTM-based Baseline and CCU-MaxNet, respectively, proving its rationality and effectiveness in cross-domain classification of MSRS data.

3) Analysis of Skip Connection: The skip connection of the memory state is designed to guide CCU-MaxNet for feature extraction and domain-invariant representation of MSRS data.

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TABLE VI
COMPLEXITY ANALYSIS OF DIFFERENT COMPARISON METHODS ON THE FOUR DATASETS

Methods	H13-H18 Dataset						
	OA (%)	κ (%)	Te_Time (s)	Model Memory (MB)	Parameters (K)	Flops (M)	GPU Memory (GB)
SCLUDA	70.14	56.95	5.28	6.73	258.194	26.076	1.4
MSDA	77.70	65.82	7.54	4.63	362.719	31.774	1.3
MLUDA	78.29	66.84	6.69	8.08	611.714	32.679	1.4
FrF-SSDA	78.05	65.03	4.27	31.50	8267.271	22.147	1.5
MDA-NET	78.88	66.13	6.52	–	558.268	19.681	11.4
SASS	77.32	65.82	7.74	6.95	692.064	88.907	1.8
HighDAN	80.17	68.49	2.14	106.70	27580.714	7330.1380	3.9
HMDA-DANet	82.38	71.43	2.81	124.98	14804.589	60633.017	7.5
Proposed (w/o FG)	82.85	72.64	2.58	97.46	11802.828	10519.957	5.3
Proposed (w FG)	84.07	74.47	5.83	97.46	11884.748	10587.066	5.4
Methods	NA-HA Dataset						
	OA (%)	κ (%)	Te_Time (s)	Model Memory (MB)	Parameters (K)	Flops (M)	GPU Memory (GB)
SCLUDA	90.00	86.30	3.04	7.31	411.842	19.029	1.6
MSDA	92.91	90.14	5.46	5.41	568.639	12.517	1.1
MLUDA	92.33	89.50	4.89	8.66	765.362	14.9010	1.4
FrF-SSDA	91.13	87.54	4.43	31.70	8292.615	25.175	1.3
MDA-NET	93.25	90.45	17.65	–	1561.948	435.300	11.5
SASS	94.53	92.46	6.01	3.83	780.834	73.780	1.7
HighDAN	94.28	92.00	4.03	106.85	27581.862	7332.6290	3.9
HMDA-DANet	95.33	93.50	4.86	125.13	14842.605	60788.730	7.5
Proposed (w/o FG)	95.74	94.00	4.79	99.10	12039.636	11491.004	5.4
Proposed (w/ FG)	96.50	95.08	11.17	99.10	12121.556	11558.113	5.5
Methods	TO-MU Dataset						
	OA (%)	κ (%)	Te_Time (s)	Model Memory (MB)	Parameters (K)	Flops (M)	GPU Memory (GB)
SCLUDA	71.63	46.09	3.36	6.77	271.006	17.773	1.4
MSDA	76.41	54.21	5.11	4.69	380.515	17.698	1.2
MLUDA	74.78	56.14	4.49	8.12	624.526	19.296	1.3
FrF-SSDA	72.18	52.73	2.80	31.50	8269.171	22.422	1.4
MDA-NET	80.05	62.38	5.82	–	590.648	21.558	11.5
SASS	80.19	63.61	2.79	6.98	699.618	12.448	1.7
HighDAN	78.67	61.47	0.96	106.71	27581.862	7332.6290	3.9
HMDA-DANet	81.56	64.23	2.24	124.98	14801.889	60642.716	7.5
Proposed (w/o FG)	82.14	69.87	1.45	97.59	11823.840	10606.137	5.3
Proposed (w/ FG)	83.84	72.31	2.53	97.59	11905.760	10673.245	5.4
Methods	H13-TO Dataset						
	OA (%)	κ (%)	Te_Time (s)	Model Memory (MB)	Parameters (K)	Flops (M)	GPU Memory (GB)
SCLUDA	89.30	78.61	1.94	6.84	289.943	20.357	1.4
MSDA	89.93	79.89	2.79	4.79	405.754	7.069	1.2
MLUDA	92.00	84.21	2.42	8.20	624.526	21.933	1.3
FrF-SSDA	95.13	90.36	2.30	31.62	8272.728	22.835	1.4
MDA-NET	91.01	82.12	4.96	–	590.648	24.701	11.5
SASS	95.39	90.96	2.83	6.98	711.852	65.486	1.7
HighDAN	93.13	86.40	1.11	106.71	27581.862	7338.199	3.9
HMDA-DANet	96.34	92.84	2.79	125.00	14808.612	60665.064	7.5
Proposed (w/o FG)	96.88	93.88	1.51	97.78	11856.261	10739.076	5.3
Proposed (w/ FG)	97.46	95.01	2.95	97.78	11938.181	10806.185	5.4

According to part (d) and the last line of part (b) in Table V, it improves the OA and κ values of CCU-MaxNet by at least 0.62% and 1.71%, respectively. These results verify the advantages of memory state in improving the cross-domain classification accuracy of the proposed CCU-MaxNet.

E. Complexity Analysis

The fourth experiment is built for a detailed complexity analysis of the proposed CCU-MaxNet framework.

1) Complexity Analysis Between Different Methods: The complexity of different methods is first analyzed, with the average testing time (Te_Time) (s), model memory usage (MB), parameters (K), Flops (M), and GPU memory usage (MB) reported in Table VI. Compared with the patch-based SCLUDA, MSDA, MLUDA, FrF-SSDA, MDA-NET, and SASS, the classification performance and Te_Time of our CCU-MaxNet on the H13–H18, TO–MU, and H13–TO datasets have absolute advantages; however, the huge difference in input size results in it having larger values on the other four indicators, especially for the NA-HA dataset with more spectral bands. Compared with the patch-free HighDAN and HMDA-DANet, CCU-MaxNet has no advantage in Te_Time due to the use of FrFT, but shows significant superiority in the other five metrics (It is worth noting that HighDAN originally has higher complexity; however, its input size is changed from 128 $\times$ 128 to 64 $\times$ 64 for the sake of comparison fairness in all experiments). Interestingly, when CCU-MaxNet is in high-efficiency mode by removing FrFT, although its classification accuracy decreases, its advantages in other metrics will be more obvious.

2) Complexity Analysis Between Different DA Methods: The complexity between our MCC-DANet and other existing methods is further analyzed, including without DA (w/o DA), vanilla adversarial discriminative DA (ADDA) [22], attention-based AADA [33], and dual-discriminator-based DA (DDA) [39]. Seven metrics are used for accuracy and complexity analysis in Table VII, containing OA, κ, Te_Time, model memory, parameters, Flops, and GPU memory.

As shown in parts (a), (c), (e), and (g) of Table VII, MCC-DANet outperforms other DA methods, while only introducing acceptable complexity mainly caused by two additional decoders. For example, for the H13–H18 dataset, compared with the other three DA methods, MCC-DANet improves the classification accuracy of the vast majority of

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TABLE VII

ABLATION STUDY OF VARIOUS DA METHODS AND VARIOUS MODULES OF THE SAME TYPE ON THE FOUR DATASETS

	Methods					H13-H18 Dataset						
	w/o DA	ADDA	AADA	DDA	Proposed	OA (%)	$\kappa \times 100$ (%)	Te_Time (s)	Memory (MB)	Parameter (K)	Flops (M)	GPU Memory (GB)
(a)	✔	✘	✘	✘	✘	13.35	—	—	—	—	—	—
	✘	✔	✘	✘	✘	79.78	67.58	5.82	59.61	8598.665	9366.932	2.8
	✘	✘	✔	✘	✘	80.26	67.95	5.83	60.29	8775.034	9389.214	2.9
	✘	✘	✘	✔	✘	82.44	71.88	5.82	60.62	8861.257	9400.716	2.9
	LSTM	LSTM2D	LSTM3D	SwinT								
(b)	✔	✘	✘	✘	✘	77.57	64.67	9.96	133.00	12658.700	12737.348	5.8
	✘	✔	✘	✘	✘	81.13	69.70	4.40	134.76	11326.860	18264.916	4.8
	✘	✘	✔	✘	✘	81.74	71.04	4.50	254.38	43572.108	60090.515	5.8
	✘	✘	✘	✔	✘	82.57	72.29	5.82	29.00	7916.940	10961.656	5.6
	✘	✘	✘	✘	✔	84.07	74.47	5.83	97.46	11884.748	10587.066	5.4
	Methods					NA-HA Dataset						
	w/o DA	ADDA	AADA	DDA	Proposed	OA (%)	$\kappa \times 100$ (%)	Te_Time (s)	Memory (MB)	Parameter (K)	Flops (M)	GPU Memory (GB)
(c)	✔	✘	✘	✘	✘	25.13	—	—	—	—	—	—
	✘	✔	✘	✘	✘	92.98	90.16	11.16	59.90	8674.697	9678.359	2.9
	✘	✘	✔	✘	✘	94.37	92.11	11.17	60.58	8851.066	9700.641	2.9
	✘	✘	✘	✔	✘	95.37	93.49	11.17	60.91	8937.289	9712.143	2.9
	LSTM	LSTM2D	LSTM3D	SwinT								
(d)	✔	✘	✘	✘	✘	91.79	88.38	19.18	134.64	12895.508	13708.394	5.8
	✘	✔	✘	✘	✘	94.58	92.37	8.42	136.40	11563.668	19235.963	5.0
	✘	✘	✔	✘	✘	95.00	92.96	8.56	256.03	43808.916	61061.562	6.0
	✘	✘	✘	✔	✘	95.33	93.45	11.16	30.64	8153.748	11932.703	5.7
	✘	✘	✘	✘	✔	96.50	95.08	11.18	99.10	12121.556	11558.113	5.5
	Methods					TO-MU Dataset						
	w/o DA	ADDA	AADA	DDA	Proposed	OA (%)	$\kappa \times 100$ (%)	Te_Time (s)	Memory (MB)	Parameter (K)	Flops (M)	GPU Memory (GB)
(e)	✔	✘	✘	✘	✘	33.61	—	—	—	—	—	—
	✘	✔	✘	✘	✘	79.02	64.79	2.50	59.64	8605.061	9393.146	2.8
	✘	✘	✔	✘	✘	81.07	67.65	2.51	60.31	8780.406	9415.429	2.9
	✘	✘	✘	✔	✘	82.14	69.23	2.52	60.62	8862.013	9424.571	2.9
	LSTM	LSTM2D	LSTM3D	SwinT								
(f)	✔	✘	✘	✘	✘	72.14	57.57	3.95	133.13	12679.712	12823.527	5.7
	✘	✔	✘	✘	✘	75.73	59.78	1.86	134.88	11347.872	18351.096	4.8
	✘	✘	✔	✘	✘	78.53	63.64	1.95	254.51	43593.120	60176.695	5.9
	✘	✘	✘	✔	✘	81.00	67.04	2.52	29.12	7937.952	11047.836	5.6
	✘	✘	✘	✘	✔	83.84	72.31	2.53	97.59	11905.760	10673.245	5.4
	Methods					H13-TO Dataset						
	w/o DA	ADDA	AADA	DDA	Proposed	OA (%)	$\kappa \times 100$ (%)	Te_Time (s)	Memory (MB)	Parameter (K)	Flops (M)	GPU Memory (GB)
(g)	✔	✘	✘	✘	✘	00.00	—	—	—	—	—	—
	✘	✔	✘	✘	✘	93.41	87.45	2.94	59.68	8615.558	9436.138	2.8
	✘	✘	✔	✘	✘	95.36	90.89	2.95	60.35	8791.159	9458.420	2.8
	✘	✘	✘	✔	✘	96.43	92.95	2.95	60.67	8873.920	9468.152	2.9
	LSTM	LSTM2D	LSTM3D	SwinT								
(h)	✔	✘	✘	✘	✘	91.28	83.61	4.93	133.32	12712.133	12956.467	5.7
	✘	✔	✘	✘	✘	93.02	86.35	2.10	135.08	11380.293	18484.036	4.8
	✘	✘	✔	✘	✘	95.06	90.37	2.25	254.71	43625.541	60309.635	5.9
	✘	✘	✘	✔	✘	96.42	92.98	2.93	29.32	7970.373	11180.776	5.6
	✘	✘	✘	✘	✔	97.46	95.01	2.95	97.78	11938.181	10806.185	5.4

land covers and obtains at least 1.63% gains in OA value while increasing the model memory usage by 37.85 MB at most. The main reason is that by introducing a new agent domain, it avoids direct alignment of MSRS data between domains with strong spectral shift and distribution discrepancy and reduces their alignment difficulty, which is different from existing methods for cross-domain MSRS data classification.

F. Necessity and Effectiveness Analysis of xMLSTM

The fifth experiment is built to verify the necessity and effectiveness of xMLSTM by replacing it with some similar modules, including LSTM [45], ConvLSTM2D (LSTM2D) [46], ConvLSTM3D (LSTM3D) [13], and SwinT [52], all of which are renowned for their expertise in modeling correlations of sequence data. The results are reported in part of (b), (d), (f), and (h) of Table VII. To maintain structure consistency, the dimension time_step in the proposed (w/ LSTM), proposed (w/ LSTM2D), and proposed (w/ LSTM3D) is fixed as the channel number, 1, and 1, whose structure settings are referenced from [13], [52], and [65], to simplify their analysis complexity, respectively.

ConvLSTM2D and ConvLSTM3D are the 2-D and 3-D versions of LSTM to explore spatial structure and spatiotemporal or spatial–spectral information while modeling long-term correlations of data [13], [46], respectively. Compared with them, the proposed (w/ xMLSTM) has obvious superiority in complexity and classification performance for MSRS data under the same number of layers. Taking the H13-H18 dataset as an example. It increases OA and κ values by at least 2.33% and 3.43%, while increasing Te_Time by only 1.43 s at most. Moreover, the structural characteristic of LSTM makes the dimension time_step of the proposed (w/ LSTM) much larger than other modules, resulting in an increase in testing time and GPU memory usage. In particular, the new multimodal component (xMLSTM) is lower than the above three modules in terms of model memory usage, parameters, and Flops.

SwinT is one of the commonly used variants of the transformer, with stronger processing capabilities for sequence data [52]. Drawing on SwinT, our xMLSTM not only has strong global modeling ability but also fully explores the relationships cross modalities and cross layers by introducing the idea of interlayer correlation modeling, extracting more representative features from MSRS data. For example, for the NA-HA

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dataset, although using SwinT instead of xMLSTM reduces the model memory usage and parameters of the whole model, it causes the loss of classification accuracy at the cost, with OA and κ values decreasing by 1.17% and 1.63%, respectively. Similar conclusions can also be drawn from the other three datasets.

As shown above, the designed xMLSTM module improves cross-domain performance without increasing too much complexity, mainly because its intermodal and interlayer correlation modeling designs can be integrated into the forward propagation process of the neural network, without affecting the efficiency of the entire framework (the high testing time is actually introduced by FrFT in the CMDC block). Furthermore, the long-term correlation learning across modalities and across layers, as well as the cross-modal difference representation, to some extent solves the problems of information redundancy and feature imbalance caused by sample imbalance, which may provide the possibility for CCU-MaxNet to explore more discriminative and domain-invariant representations of MSRS data. These experimental results verify the rationality and effectiveness of xMLSTM, which lays the foundation of the proposed CCU-MaxNet framework.

V. Conclusion

In this article, a new CCU-MaxNet framework has been proposed for UDA in end-to-end classification of MSRS data. An xMLSTM cell with memory state is first developed as a specialized multimodal deep learning component to build two plug-and-play blocks (i.e., down-block and up-block), providing an effective way to model long-term dependencies between modalities and between hierarchical features. Then, a CMDC block is designed to enhance the complementary modality-specific information. Using these blocks as basic components, U-MaxNet is built to fully explore the representation advantages of MSRS data, extracting complementary enhanced fusion features. MCC-DANet is further developed to align the multimodal domain distributions at both pixel and semantic levels, thus extracting domain-invariant multimodal features from MSRS data for their cross-domain classification. Moreover, the two working modes (high-precision mode and high-efficiency mode) also make the proposed CCU-MaxNet framework more flexible and effective for practical scenarios and tasks. Extensive experiments on four benchmark datasets confirm that CCU-MaxNet outperforms other state-of-the-art methods in terms of both quantitative metrics and visual quality on the task of cross-domain MSRS data classification.

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